

SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning constitutes a core faculty of human cognition, yet it remains an unresolved challenge for contemporary vision–language models (VLMs). Prior evaluation resources have predominantly been instantiated on synthetic or LLM-generated environments constrained to a narrow task repertoire and puzzle-like configurations, and therefore fail to reproduce the scene complexity, visual noise, and heterogeneous spatial relations characteristic of natural imagery. To close this gap, we introduce **SPATIALAB**, a comprehensive benchmark for quantifying the spatial reasoning competence of VLMs under realistic, unconstrained visual conditions. SPATIALAB consists of 1,400 image–question–answer triples partitioned into six top-level categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each decomposed into five subcategories, yielding a taxonomy of 30 distinct task types. Every subcategory is populated with at least 25 items and every top-level category with at least 200 items, and each item is instantiated in both a multiple-choice and an open-ended evaluation format. A controlled evaluation spanning a diverse set of state-of-the-art VLMs, comprising open- and closed-source models, reasoning-optimized systems, and spatially specialized architectures, quantifies a substantial competence gap relative to human annotators. Under the multiple-choice protocol, InternVL3.5-72B attains the highest model accuracy at 54.93%, against a human accuracy of 87.57%. Under the open-ended protocol, every model incurs an additional accuracy reduction of approximately 10–25 percentage points, with GPT-5-mini attaining the maximum at 40.93% against a human accuracy of 64.93%. These measurements localize systematic deficits in resolving compositional spatial relations, depth ordering, navigation, and 3D geometry. By instantiating a diverse, real-world evaluation protocol, SPATIALAB isolates the principal failure modes and improvement axes for VLM spatial reasoning and provides a reproducible instrument for advancing robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a core component of human cognition, comprising the interpretation of spatial layouts, the mental construction and manipulation of object configurations, and the planning of trajectories through an environment. It underpins application domains that range from robotics, autonomous driving, and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents, that is, autonomous systems that perceive and act within dynamic environments, reliable spatial competence entails inferring geometric relations, depth cues, occlusion structure, and agent-centric reference frames from scenes that are visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models have substantially improved multimodal representation learning and language grounding, yet their spatial inferences remain unreliable under realistic imaging conditions (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks, however, are narrow in scope and structurally simplified. The majority target elementary sub-problems, such as binary spatial-relation classification, coarsely discretized depth categories, or schematic navigation cues, and are typically instantiated in synthetic or otherwise

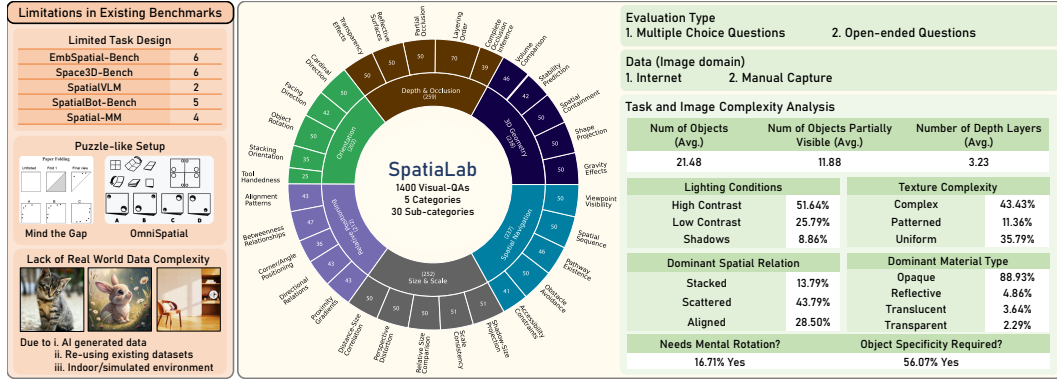


Figure 1: **Overview of SPATIALAB.** The benchmark is designed to remedy the limitations of prior datasets (left), instantiates 1,400 visual QA items spanning 5 categories and 30 subcategories (center), and supports systematic evaluation under both multiple-choice and open-ended protocols. The corpus is characterized by high task and image complexity, controlled for object count, depth layering, lighting, texture, relational structure, and material composition (right).

controlled settings that suppress visual clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Such controlled protocols attenuate the perceptual and inferential load and thereby induce apparent metric saturation, while concealing failure under distribution shift. Consequently, capabilities that are central to real-world competence, including robust occlusion inference, scale-consistent estimation across viewpoints, and planning under partial observability, remain systematically under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). As a result, models that report high accuracy on existing benchmarks frequently degrade on integrated, multi-step inference, which constrains safe deployment in real-world environments where spatial errors incur practical cost.

A rigorous evaluation must therefore jointly cover all principal axes of spatial reasoning: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. In contrast to current task-isolated protocols, human cognition integrates these axes jointly under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). A robust benchmark should consequently satisfy three design requirements: it should evaluate both discriminative selection and free-form generation, calibrate model performance against a human reference, and source imagery from natural scenes that surface the failure modes suppressed by simulator-derived datasets. These requirements motivate the design of our benchmark.

To satisfy the requirements above, we introduce **SPATIALAB**, a benchmark constructed to quantify the spatial reasoning competence of vision-language models under realistic, unconstrained visual conditions, as summarized in Figure 1. **SPATIALAB** provides a class-balanced suite of image-question-answer items spanning a broad taxonomy of spatial tasks and instantiates each item under both multiple-choice and open-ended evaluation protocols. The benchmark is designed to maximize photographic diversity, to include compositional relational queries, and to require the joint integration of depth, perspective, and 3D structure. We evaluate a broad set of state-of-the-art VLMs, comprising open- and closed-source models and reasoning-optimized systems, calibrate them against a human reference, and conduct a systematic error analysis. Our principal contributions are as follows:

1. We introduce **SPATIALAB**, a large-scale benchmark of 1,400 image-question-answer items spanning six principal categories and thirty task types under realistic, unconstrained conditions. With a lower bound of 200 items per category and 25 per subcategory, the corpus provides class-balanced coverage and instantiates every item under both a multiple-choice (SPATIALAB-MCQ) and an open-ended (SPATIALAB-OPEN) protocol.
2. We conduct a **systematic evaluation** of more than 25 state-of-the-art vision-language models, spanning open- and closed-source, reasoning-oriented, and spatially specialized

Table 1: Comparison of spatial reasoning benchmarks along task setup, data properties, evaluation protocol, and best reported model performance.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval		A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth				
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M,E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93

architectures, against human references, and quantify a consistent and statistically substantial spatial-reasoning gap.

3. We provide **fine-grained error analyses and diagnostic studies** that localize systematic failure modes in depth estimation, navigation, occlusion inference, and 3D geometry, and translate them into concrete, actionable directions for robust, human-aligned spatial understanding.
4. We evaluate multiple intervention strategies for improving VLM spatial reasoning, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and a **multi-agent architecture (SPATIOXOLVER) performing multi-step reasoning**, and characterize their gains and limitations across the multiple-choice and open-ended protocols.

Under the multiple-choice protocol, the strongest models attain approximately 50–55% accuracy, whereas human annotators exceed 85%. Under open-ended generation, model accuracy decreases by a further 10–25 percentage points, with the best systems near 40% against approximately 65% for humans. The error analysis localizes concentrated deficits in occlusion inference, scale consistency, and multi-step navigational planning, indicating limited compositional spatial representations and inference. Our study further characterizes which interventions yield robust gains, which amplify pre-existing biases, and the determinant role of perceptual grounding for generalizable spatial reasoning. By providing the task suite, baseline measurements, and diagnostic analyses, SPATIALAB establishes a reproducible foundation for progress and comparison in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMS

Visual-spatial reasoning, the capacity to perceive, represent, and manipulate the spatial relations among objects, is a central component of human cognition and a prerequisite for embodied agents that navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we model a scene as a set of objects $O = \{o_1, \dots, o_n\}$, each endowed with attributes (e.g., position, orientation, size) expressed in a world coordinate system W . Visual-spatial reasoning is then formalized as the estimation of a mapping $f : (I, Q) \mapsto R$, where I denotes an image (or image sequence), Q denotes a natural-language query, and R denotes a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ over predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), estimating f is difficult because it requires coupling noisy visual perception with compositional linguistic abstraction subject to embodied constraints, including perspective projection, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early treatments of spatial reasoning in VLMs have largely been confined to elementary spatial relations or narrow task specifications (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), frequently instantiated on synthetic datasets such as CLEVR (Johnson et al., 2016), on structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or on puzzle-like simulated protocols such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis

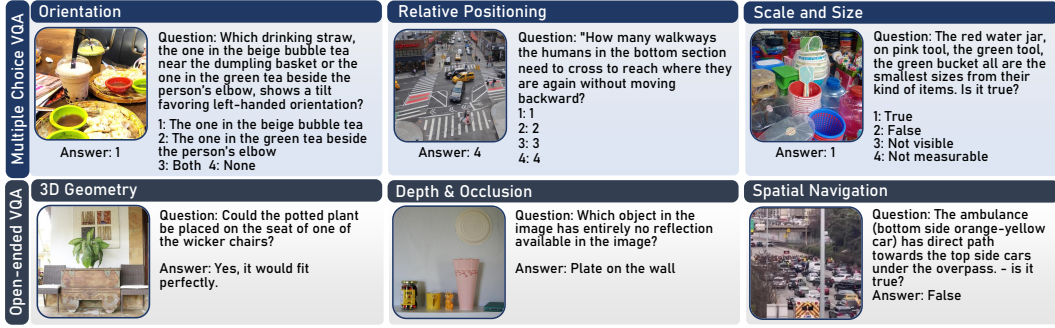


Figure 2: Representative items from the six categories, shown under both the open-ended and MCQ protocols.

et al., 2025), as summarized in Table 1¹. Although effective for controlled evaluation, these protocols oversimplify the underlying spatial structure, suppress environmental noise, and omit dynamic factors such as temporal variation and viewpoint change. To overcome these limitations, we contend that spatial reasoning must be benchmarked in rich, real-world contexts that reproduce scene complexity, temporal variation, and multiple viewpoints. Such conditions subject VLMs to the challenges they would encounter in embodied or interactive deployments, including robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

In SPATIALAB, spatial reasoning is partitioned into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each mapped to a distinct cognitive faculty characterized in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is further decomposed into five subcategories, yielding a taxonomy of thirty task types that isolate fine-grained reasoning primitives, such as directional alignment, occlusion inference, and stability prediction. This taxonomy is motivated by the requirement to cover the full space of visual-spatial reasoning challenges. By decomposing complex reasoning into measurable and independently testable units, the evaluation quantifies not only whether a model solves a spatial task in aggregate but also which specific reasoning dimensions remain fragile or under-developed (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

The benchmark further adopts two complementary evaluation protocols: multiple-choice and open-ended question answering. The multiple-choice protocol measures recognition and fine-grained discrimination among distractors, whereas the open-ended protocol measures generative inference and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Jointly, the two protocols instantiate real-world reasoning demands and expose complementary strengths and weaknesses in model behavior. By combining a principled taxonomy, diverse real-world imagery, and a dual-protocol evaluation, SPATIALAB constitutes a structured framework for systematic benchmarking. This design supports fine-grained model comparison, calibration against human-level performance, and precise characterization of the mechanisms underlying robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains an open problem for vision-language models (VLMs), particularly under naturalistic conditions in which occlusion, perspective distortion, and cluttered backgrounds co-occur. To address this, we present **SPATIALAB**, a benchmark constructed to systematically quantify VLM competence in complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 image-question-answer items partitioned into six top-level categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each further decomposed into five subcategories, yielding 30 distinct task types. Every subcategory contains at

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

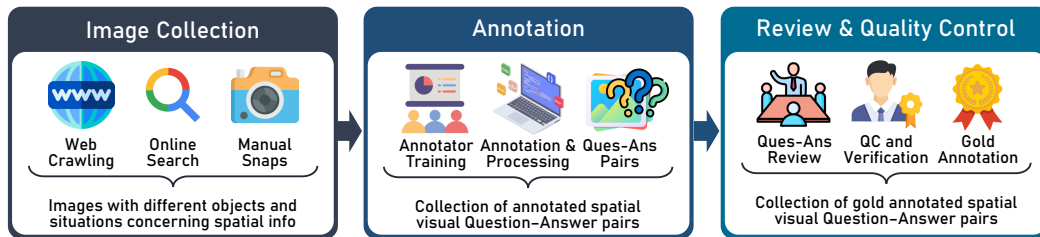


Figure 3: **Data construction pipeline for SPATIALAB.** Images are acquired via web crawling, targeted retrieval, and manual capture, and are subsequently annotated with structured spatial question-answer pairs.

least 25 items and every top-level category at least 200, enforcing class-balanced coverage across the taxonomy. **SPATIALAB** instantiates each item under both a multiple-choice and an open-ended protocol, thereby measuring discriminative accuracy jointly with generative inference and explanatory competence, consistent with prevailing multimodal evaluation practice while targeting spatial-specific reasoning. High visual diversity, fine-grained annotation, and a rigorous quality-control pipeline render **SPATIALAB** a reliable and scalable testbed; Figure 2 presents representative items from the six categories under both protocols, illustrating the diversity and difficulty of real-world spatial reasoning.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was assembled using three complementary acquisition strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manually captured photographs from diverse indoor and outdoor environments (Figure 3). This multi-source pipeline induces diversity across object categories, environmental conditions, and spatial configurations. To control for natural variability explicitly, each image was annotated along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Every dimension is represented in the final corpus, ensuring that the benchmark spans a broad range of real-world, in-the-wild conditions rather than simplified or synthetic configurations. A quantitative complexity analysis of the corpus yields, on average, 21.48 objects per image, of which 11.88 are partially visible, distributed over 3.23 distinct depth layers. Spatial reference chains (e.g., “object A is left of B, which is behind C”) comprise 2.07 links on average, evidencing multi-step relational structure. In addition, 16.71% of the items require mental rotation and 56.07% require object specificity (a designated color, pattern, or other distinguishing attribute). These statistics establish that **SPATIALAB** captures cluttered, depth-layered, and relationally dense scenes, which are characteristic of real-world spatial reasoning yet under-represented in synthetic datasets.

Annotation. Annotation was conducted in three sequential phases. In the first phase, annotators completed a targeted training protocol (detailed in Appendix C.2) to standardize the interpretation of spatial categories and subcategories. In the second phase, each image was paired with one or more spatial QA tasks spanning a range of reasoning types, including proximity gradients, alignment patterns, occlusion inference, and stability prediction; questions were formulated to balance perceptual grounding against higher-order inference (e.g., predicting stability under gravity constraints). In the third phase, each QA pair was encoded in both a multiple-choice (four-option) and an open-ended format to ensure comparability across the two evaluation protocols. This procedure produced approximately 1,500 spatial visual QA pairs across the six categories and 30 subcategories.

Review and Quality Control. To ensure reliability, all annotation outputs were processed through a three-tier review and validation pipeline. In the first stage, each QA item was checked for semantic validity and consistency with its designated subcategory. In the second stage, independent annotators verified the correctness of both the question phrasing and the designated answer, with an emphasis on removing ambiguity and enforcing task clarity. In the third stage, a gold-standard annotation round established a set of high-confidence QA pairs for evaluation. All image metadata was stripped, and the corpus was curated to mitigate copyright concerns, although minor oversights may remain. This pipeline yielded a final benchmark of 1,400 validated QA items.

Table 2: Multiple-choice accuracy (%) ($\uparrow$) on SPATIALAB-MCQ, reported per question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

4 EXPERIMENTS AND EVALUATION

Models. We partition the evaluated models into four groups. The **proprietary** group comprises GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). The **open-source** group comprises InternVL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). The **reasoning** group comprises o4-mini (OpenAI, 2025), the Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). The **spatial reasoning specialist** group comprises SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we restrict evaluation to the open-ended protocol: these models are not instruction-tuned and were trained predominantly on open-ended objectives, which induced an excessive refusal rate under the MCQ protocol. To facilitate comparison in the results tables, we apply the following color encoding: **proprietary**, **open-source up to 7B**, **larger than 7B**, **proprietary reasoning**, **open-source reasoning**, **spatial specialists**, and **human baseline**.

Evaluation Protocol and Metrics. We adopt accuracy as the primary metric under both the MCQ and open-ended protocols. Under the **MCQ protocol**, we use direct prompting in which the model returns the index of its selected option, and each response is scored automatically against the ground-truth option. The exact prompts and configuration are specified in Appendix D.1. Under the **open-ended protocol**, the model is prompted to generate a free-form answer, which is then scored by a large-language-model judge. The corresponding prompts and configuration are given in Appendix D.2. To validate the judge, we additionally quantified LLM-human agreement, as detailed in Appendix D.2. Table 5 reports that the LLM judge (Gemini-2.5-Flash) attains substantial agreement with human annotators, with a Cohen’s kappa of 0.738 against the majority vote and in the range 0.681–0.795 against individual annotators. Raw accuracy against the majority vote is 0.880, and the Fleiss’ kappa

Table 3: Open-ended accuracy (%) (↑) on SPATIALAB-OPEN, reported per question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

among human annotators is 0.774, indicating strong inter-rater reliability. Additional specifications of the experimental setup, prompts, and scoring procedures are provided in Appendix D.

Interventions for Improving Visual Reasoning. To improve spatial reasoning on the benchmark, we evaluate a set of intervention strategies: the models’ inherent reasoning mechanisms (Appendix H.1), Chain-of-Thought (CoT) prompting (Appendix H.2), CoT augmented with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and a multi-agent system (Appendix H.5). For SFT specifically, Qwen-VL2.5-3B-Instruct was fine-tuned on a stratified 40% partition of the dataset, improving generalization on relational and geometric tasks. Evaluation on the held-out 60% partition exhibited reduced error rates in navigation and orientation, with additive gains when CoT and SFT were combined. The agentic condition, by contrast, showed that although orientation improves substantially, the remaining categories stagnate or degrade, indicating that deeper perceptual and spatial grounding is required beyond multi-step reasoning.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports model accuracy under the multiple-choice protocol on SPATIALAB-MCQ. Accuracy varies substantially across models, tasks, and model families: overall model accuracy spans approximately 30–55% (chance-level accuracy is 25%), whereas the human reference attains 87.57%, indicating a large residual gap. The best-performing model is the open-source InternVL3.5-72B at 54.93% accuracy. Reasoning- and instruction-tuned models (e.g., o4-mini at 53.21%, Gemini-2.5-Flash-Thinking at 52.93%, GPT-5-mini at 54.29%) and several very large open-source models (e.g., InternVL3.5-72B at 54.93%, Llama-3.2-90B at 50.36%) occupy the top of the ranking, but parameter count alone is not determinative: Llama-3.2-11B, for example, attains only 30.50% despite its scale. At the category level, Orientation and 3D Geometry are the categories on which strong models perform best (several models exceed 60% on one or both), whereas Spatial Navigation, Depth & Occlusion, and Size & Scale exhibit higher variance and are systematically harder. Notably, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) attain

only low- to mid-40% accuracy overall, indicating that architectural specialization does not automatically yield higher MCQ accuracy across the diverse item set. Depth & Occlusion and Size & Scale yield heterogeneous results, with some architectures resolving occlusion relatively well while others fail, consistent with differing inductive biases or training distributions. These patterns indicate that instruction-tuning and explicit reasoning capacity selectively benefit geometric and orientation reasoning, while multi-step grounding tasks such as navigation remain an unresolved weakness.

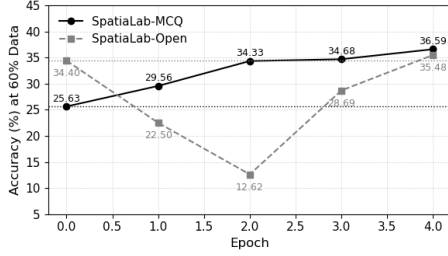
SPATIALAB-OPEN. The open-ended accuracies reported in Table 3 are substantially lower than the corresponding MCQ accuracies and exhibit a wide range: overall accuracy spans approximately 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini), against a human reference of 64.9%, indicating a large residual gap. Proprietary models generally lead under the open-ended protocol (e.g., GPT-5-mini at 40.93%, GPT-4o-mini at 26.00%, Gemini-2.5-Flash at 30.93%); however, reasoning-tuned variants are particularly prominent, with o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) outperforming many open-source counterparts, consistent with instruction/CoT-style tuning substantially improving generative outputs. Most small open-source models cluster at the lower end (many below 20%; InternVL3.5-1B at 9.64%, Qwen-VL2.5-3B at 12.93%), whereas several very large open models reach the low-20% range (e.g., InternVL3.5-72B at 23.36%, Qwen-VL2.5-72B at 24.64%), indicating that scale alone confers only modest gains. The spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) attain only 10–13% overall, indicating that architectural or task specialization does not automatically yield robust open-ended generation across the diverse item set. At the subtask level, the strongest models perform comparatively better on orientation and on selected relative-position items (e.g., GPT-5-mini at 49.53% on relative position and 37.13% on orientation), whereas depth & occlusion, size & scale, and especially spatial navigation remain consistently difficult (many models score below 30% on these subtasks). This pattern corroborates the earlier finding that multi-step grounding and perceptual inference are the principal bottlenecks for open-ended generative answers.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

Aggregated over 25 models, the mean MCQ→open-ended accuracy gap on SPATIALAB is 23.0% ($\sigma = 5.5\%$), with per-subtask means ranging from 22.89% (spatial navigation) to 24.57% (3D geometry). Specialist spatial reasoning models exhibit the largest gaps (approximately 27%), concentrated in spatial navigation (up to 36.68%) and orientation (34.44%), whereas reasoning-oriented models yield smaller gaps (approximately 19%) and lower subtask variance, consistent with a stabilizing effect of instruction-tuning and CoT decoding on generative outputs. Correlation analysis confirms that the spatial-navigation gap is the dominant predictor of the overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) as secondary contributors. Negative or near-zero gaps (e.g., -1.98 for Llama-3.2-11B in depth & occlusion; -3.18 for o4-mini in relative position) indicate that poorly calibrated MCQ distractors can misrepresent the underlying competence. We attribute this format sensitivity to four interacting factors: the structural advantage of the constrained MCQ output space, a specialization bias toward categorical tasks, the sequential-reasoning demands of spatial navigation, and calibration differences in generative output. These results imply that the MCQ protocol alone overestimates practical spatial reasoning competence. We therefore recommend complementing MCQ with open-ended evaluation, auditing distractors, adopting stepwise generation or instruction-tuning, and reporting per-subtask diagnostics in order to recover the true model competence and suppress format-dependent artifacts. Further detail is provided in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. Our analysis (Appendix E) indicates that VLMs attain selective peaks but lack uniform spatial competence. Closed-source models reach the highest subtask accuracies (e.g., 85.71% on *Stacking Orientation*) yet degrade to near-chance on *Relative Size Comparison*. Scaling of open-source models raises the achievable ceiling (80.56% on *Corner/Angle Positioning*) but does not eliminate catastrophic failures (2.0% on *Object Rotation*). Reasoning-augmented and spatially specialized models contribute localized gains, for instance in occlusion inference and navigation, yet remain capped below 55% on many physical-abstraction tasks (*Gravity Effects*, *Stability Prediction*). The subtask error distributions further localize systematic deficits in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cue



(a) SFT accuracy trajectory as a function of training epoch.

(b) Per-category accuracy before and after SFT, under the MCQ and open-ended protocols, with the corresponding gains.

Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

Figure 4: SFT results: (a) the accuracy trajectory over training epochs, and (b) the final per-category accuracies.

integration (*Reflective Surfaces*), whereas comparatively strong results on *Obstacle Avoidance* are consistent with shortcut exploitation rather than robust planning. In aggregate, these patterns indicate that current VLMs rely predominantly on surface-level correlations and lack stable internal encodings of orientation, physics, and compositional logic. Progress will likely require both richer training distributions (e.g., embodied and physics-driven data) and architectural mechanisms that enforce geometric and reference-frame grounding.

SPATIALAB-OPEN. The error analysis of SPATIALAB-OPEN (Appendix F) shows that even the strongest closed-source models (e.g., GPT-5-mini at 58.14% on *Directional Relations*) degrade on tasks such as *Proximity Gradients*, indicating that their advantage is task-specific rather than uniform. Large open-source models attain strong accuracy on *Depth & Occlusion* (e.g., 60.0%) yet fall to 9.3% on *Proximity*, demonstrating that scale alone does not guarantee robustness. Smaller models exhibit isolated strengths (e.g., 26.0% on *Relative Size Comparison*) but often fail completely (e.g., 0.0% on *Betweenness*), consistent with the representational limits of compact architectures. Reasoning-tuned systems show partial promise, with Gemini-2.5-Flash-Thinking reaching 75.0% on *Tool Handedness*, yet peer models drop to single digits, indicating fragile integration. Specialized spatial models are weakest overall, rarely exceeding 20% and frequently failing on core relational tasks. The failure distribution localizes systematic deficits in occlusion handling, orientation, and multi-step relational chaining, while partial success on size-scale cues is consistent with reliance on superficial correlations. These results implicate three root causes: an absence of representations for spatial analogies, architectural limits coupled to scale and training diversity, and brittle reasoning pipelines. Collectively, they motivate unified geometric encodings, physics-aware reasoning, and embodied training data as prerequisites for progressing from isolated peaks toward consistent spatial competence.

Qualitative Error Analysis. To characterize model weaknesses in detail, we conducted a qualitative error analysis (Appendix I) and found that failures partition into a small number of recurring classes rather than constituting random noise. The dominant classes are: spatial mislocalization, in which models confuse referents in crowded scenes; perspective and scale errors, in which reliance on canonical object-size priors overrides image-based cues; and occlusion or ordering errors, particularly for thin or partially hidden structures. We additionally observe attribute confusion, in which perceptual properties are conflated with functional roles, and open-ended rationalization, in which the model generates fluent but visually ungrounded narratives. These error classes are stable across architectures and prompting strategies, consistent with structural biases in current VLMs rather than dataset-specific noise. Notably, the failure rate increases sharply when multiple cues must be fused, such as jointly resolving depth ordering and relative size, and confidence calibration is markedly degraded under the open-ended protocol. Our diagnostic protocols confirm that models frequently disregard minimal but decisive visual features, rely on brittle heuristics, and fail to revise their predictions under counterfactual image edits. The root causes are attributable to insufficient object-centric binding, an absence of geometric supervision, and training objectives that reward plausibility over grounding. In aggregate, these findings indicate that current VLMs achieve strong coarse perception but degrade under multi-cue integration and grounded reasoning, motivating geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently outperform their non-reasoning baselines under both protocols, with the largest gains on relational and orientation tasks (e.g., +13.1% on Relative Positioning under MCQ). Although reasoning stabilizes open-ended performance, the gains are non-uniform across categories and are negative on Size & Scale, indicating that reasoning modules improve logical consistency but do not resolve grounding or scale sensitivity. Per-category comparisons are provided in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting yields negligible benefit and frequently reduces accuracy (Table 19), with orientation the only category exhibiting consistent gains. This indicates that step-by-step reasoning assists directional alignment but fails on tasks requiring robust perceptual grounding, namely depth, scale, and spatial navigation. In contrast to textual reasoning, CoT here amplifies flawed priors rather than correcting them, confirming that solving SPATIALAB requires perceptual grounding beyond logical chaining (Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Augmenting CoT with a self-reflection stage, in which the model is explicitly prompted to review its prior reasoning step-by-step and correct errors (see Figure 8 in the Appendix for the prompt), yields modest gains under the MCQ protocol, most pronounced on geometry and depth. These gains do not transfer to the open-ended protocol (Appendix H.3). This confirms that reflective reasoning cannot compensate for absent perceptual grounding, exposing a core limitation of current VLMs.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning of Qwen2.5-VL-3B-Instruct on a stratified 40% partition of SPATIALAB (~560 items) consistently improves MCQ accuracy across all spatial reasoning categories, but yields negligible and occasionally negative transfer to open-ended tasks. This divergence is consistent with overfitting to the task-specific answer distribution, corroborating the concern that SFT *teaches to the test* rather than inducing generalizable reasoning (Wang et al., 2025). The sharp reduction in open-ended accuracy indicates internal representations biased toward categorical discrimination and unstable under generation. A plausible mechanism is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that favor discrete-choice alignment over language-grounded spatial reasoning. In summary, SFT improves accuracy in structured formats but risks an algorithmic constraint that raises accuracy on constrained tasks while stalling progress on naturalistic spatial reasoning. Further detail is provided in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To characterize agentic reasoning, we employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) for structured spatial reasoning over images. The framework decomposes complex visual reasoning into specialized sub-tasks assigned to dedicated agents, including modules for object segmentation, attribute extraction, spatial-relation mapping, and transformation tracking. By consolidating these agent outputs into a unified structured representation, the system propagates low-level perceptual cues through higher-order relational inference. We instantiate the pipeline with Gemini-2.5-Flash at low temperature to enforce approximately deterministic outputs across the multi-stage refinement process. The results show substantial gains on the Orientation category, of +8.00% under MCQ and +36.00% under open-ended evaluation. These gains are not uniform: Depth & Occlusion and Spatial Navigation incur pronounced open-ended declines of -24.00% and -12.00%, respectively. These decrements indicate that when the underlying perceptual priors, such as depth or occlusion cues, are weak, multi-step reasoning reinforces rather than corrects misconceptions. Overall, agentic workflows improve tasks reducible to sequential alignment but cannot compensate for absent perceptual grounding in complex 3D scenes. Further detail is provided in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet most benchmarks are instantiated on synthetic scenes, templated questions, or a narrow task scope that omits real-world visual variability. We introduce SPATIALAB, a benchmark of 1,400 image-question-answer items spanning six top-level spatial categories and 30 subcategories, including 3D geometry, occlusion, and navigation. The benchmark instantiates each item under both a Multiple-Choice and an Open-Ended protocol, thereby separating option-based pattern matching from genuine generative reasoning. Its quality is enforced

by a multi-stage review pipeline and complexity controls over lighting, texture, and spatial diversity. An evaluation spanning more than 25 contemporary models and human references establishes that SPATIALAB functions as a precise diagnostic instrument for localizing the weaknesses of current vision–language systems.

The detailed error analysis localizes systematic failures in transparency handling, multi-step navigation, and occlusion inference, indicating that models rely on surface-level texture cues rather than internalized 3D geometric representations. The intervention study shows that supervised fine-tuning (SFT) produces clear gains under the structured MCQ protocol but concurrently induces instability under open-ended reasoning, consistent with catastrophic forgetting of linguistic priors when optimizing for discrete answers. Chain-of-Thought reasoning is effective on sequential tasks such as Orientation but does not correct fundamental perceptual errors in Depth and Scale, and frequently amplifies hallucinations rather than resolving them. Agentic decomposition exhibits an analogous trade-off, improving specific alignment tasks while degrading holistic scene understanding. These divergent behaviors establish that scaling and reasoning scaffolds alone are insufficient; attaining human-level spatial intelligence will require architectural mechanisms that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, particularly under visually complex, real-world conditions. **SPATIALAB** contributes 1,400 diverse QA items across six spatial categories and thirty subcategories, supporting fine-grained evaluation under both the MCQ and open-ended protocols. Despite the application of several intervention strategies, our experiments quantify substantial gaps between state-of-the-art models and human performance, with open-ended generation incurring an additional 10–25% accuracy reduction relative to multiple-choice accuracy. The error analysis localizes persistent failures in depth estimation, occlusion inference, 3D geometry, and multi-step navigation, consistent with reliance on superficial cues rather than grounded spatial representations. **SPATIALAB** exposes these limitations and provides a structured testbed for diagnosing spatial competence. The benchmark supports holistic evaluation and yields actionable directions for developing VLMs with human-aligned spatial understanding. Continued progress will require richer training signals (e.g., embodied interaction and action-conditioned feedback) together with architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We implemented several measures to ensure ethical compliance throughout this work. All image metadata was removed to protect privacy, and personally identifiable features (e.g., human faces and license plates) were minimized or excluded wherever feasible. The dataset was curated to mitigate copyright concerns, with explicit attention to sourcing and licensing, although minor oversights may persist despite these efforts. The benchmark construction procedure was designed for transparency and fairness, such that no demographic group is singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To support reproducibility, the main text and appendix specify the dataset construction procedure, the annotation pipeline, and the evaluation protocol in full. All filtering, quality-control, and validation steps are documented explicitly, together with formal definitions of the task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in the appendix to enable consistent replication of the reported results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to infer geometric structure and relational layout in visual scenes, is a central requirement for embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) probed compositional reasoning using synthetic or curated templates, which afford controlled analysis but exhibit limited ecological validity. With the emergence of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), spatial reasoning has been reassessed on more naturalistic datasets, yet systematic deficits persist. Despite strong performance on captioning and visual QA, these models degrade on tasks requiring depth ordering, occlusion inference, or consistent reference-frame alignment. This discrepancy indicates a broader property: parameter scaling alone does not guarantee progress on structured spatial understanding.

Spatial reasoning benchmarks. A range of benchmarks have recently been proposed to probe these deficits (Table 1). SCANQA (Azuma et al., 2022) was among the first embodied QA datasets requiring reasoning over 3D indoor environments, but its limited scale (800 scenes) and shallow reasoning complexity constrain generalization. VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) extended coverage to real-world and household contexts but relied on template-driven question generation, which reduces linguistic variability and over-represents simple binary or multiple-choice relations. More recent efforts, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), increased task variety yet remained constrained in scale (2.2K and 211 items, respectively) and in annotation style (template-heavy or manual but narrow). SPATIALRGPT-BENCH and BLINK-SPATIAL introduced mixed domains and up to 14 categories, improving diversity, but continued to emphasize either static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The trend toward embodiment is evident in datasets such as ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions. Its template-based annotation, however, renders the questions less natural and susceptible to heuristic shortcuts. Similarly, SPATIALVLM (Chen et al., 2024b) provides unprecedented scale (billions of QA pairs), but its two-category coverage precludes fine-grained diagnostic evaluation. At the opposite extreme, high-fidelity manual datasets such as VSI-BENCH (Yang et al., 2024) achieve higher annotation quality, but at only 288 scenes they are insufficient for training or robust cross-model comparison. With respect to difficulty, most existing benchmarks are bounded at the “medium” level (Mm in Table 1) and rarely extend to the “hard” or “extremely hard” levels that require multi-step causal inference. This leaves a substantial evaluation gap for next-generation models.

Evaluation protocols. The evaluation protocol and analysis depth are equally consequential. Most prior work employs closed-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which simplify scoring but constrain the expressivity of model outputs. A small number of datasets, including ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, introduce open-ended evaluation, but only SPATIALAB-OPEN couples it with high task complexity, manual puzzle-style annotation, and comprehensive error analysis. This combination supports a finer characterization of systematic weaknesses beyond aggregate accuracy, including calibration and cross-model divergence, which existing datasets rarely capture. Absent this granularity, reported gains can be misleading, since models may overfit to dataset-specific biases rather than acquire robust spatial competence.

Positioning of SPATIALAB. SPATIALAB is designed to remedy these limitations. In contrast to prior datasets covering between 2 and 14 task categories, SPATIALAB covers 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation, providing substantially higher task diversity. Unlike template-based benchmarks, every question is manually authored and puzzle-inspired, which increases linguistic diversity and reduces shortcut opportunities. SPATIALAB is furthermore fully embodied and multimodal, spanning both static and dynamic visual contexts, and thereby bridges the gap between robotics-oriented datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). As reported in Table 1, even the strongest baselines, InternVL3.5-72B and GPT-5-mini, remain well below 60%, and the open-ended variant yields only 40.93% accuracy. This large residual gap quantifies the difficulty of the benchmark and its utility for driving future research.

Toward next-generation evaluation. By combining high difficulty, broad coverage, manual annotation, and open-ended tasks, SPATIALAB extends the spatial-reasoning benchmark landscape from template-driven, narrow evaluation to comprehensive, embodied testing. This design supports not only measuring answer correctness but also identifying systematic error patterns, probing causal and counterfactual reasoning, and quantifying calibration reliability. In this respect, SPATIALAB provides a rigorous foundation for the next phase of multimodal research, complementing datasets such as MIND THE GAP and VLM4D that address temporal or generative aspects but remain narrower in coverage. We anticipate that future work will employ SPATIALAB as a high-bar diagnostic instrument that advances models beyond surface-level recognition toward genuinely grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we conducted a systematic examination of existing spatial reasoning benchmarks. Prior efforts have established important foundations, yet they exhibit structural and methodological constraints that limit their utility for evaluating current VLMs. These constraints fall into three classes: synthetic-domain bias, limited task scope, and restrictive evaluation formats. Table 4 summarizes the principal limitations identified.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA are instantiated on synthetic environments or template-generated linguistic structures. Although these conditions isolate compositional reasoning under controlled variation, the visual domain lacks real-world complexity and the linguistic space is narrow, so models can exploit template regularities in place of grounded spatial interpretation. Related simulated benchmarks such as ROBOSPATIAL exhibit the same limitation, yielding a large sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but are restricted to indoor scenes, precluding evaluation in the outdoor or mixed environments required for embodied agents. High-fidelity manual datasets such as VSI-Bench provide richer annotation but remain small in scale, which limits their utility for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack a detailed taxonomy, which precludes diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, depend heavily on binary or MCQ formats, which can inflate measured performance by admitting option elimination in place of generative spatial reasoning. SPATIALAB mitigates these issues by combining the MCQ and open-ended protocols over a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations identified in representative prior spatial benchmarks relative to SPATIALAB, categorized by domain constraint, evaluation format, and task taxonomy.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale ($< 1K$ scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~ 200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning entails more than object recognition; it requires inferring how objects relate, align, and interact in physical space. To probe this competence, we introduce *SpatialLab*, a benchmark constructed to evaluate fine-grained facets of spatial understanding across a diverse category set. The tasks in SpatialLab instantiate the manner in which humans describe and reason about environments, spanning relative positioning (e.g., *What’s to the left of the microwave?*) through geometric inference (e.g., *Would this stack fall if another book is added?*). Each category is decomposed into subcategories that isolate distinct but essential reasoning dimensions, including occlusion, orientation, navigation,

and physical effects such as gravity. By formulating each subtask in concrete, everyday terms, SpatiaLab requires models to move beyond object detection toward structured spatial interpretation. This design quantifies how well systems generalize across the contexts relevant to robotics, AR/VR, autonomous driving, and human–computer interaction. The following sections provide concise task-level specifications for every category and subcategory in SpatiaLab.

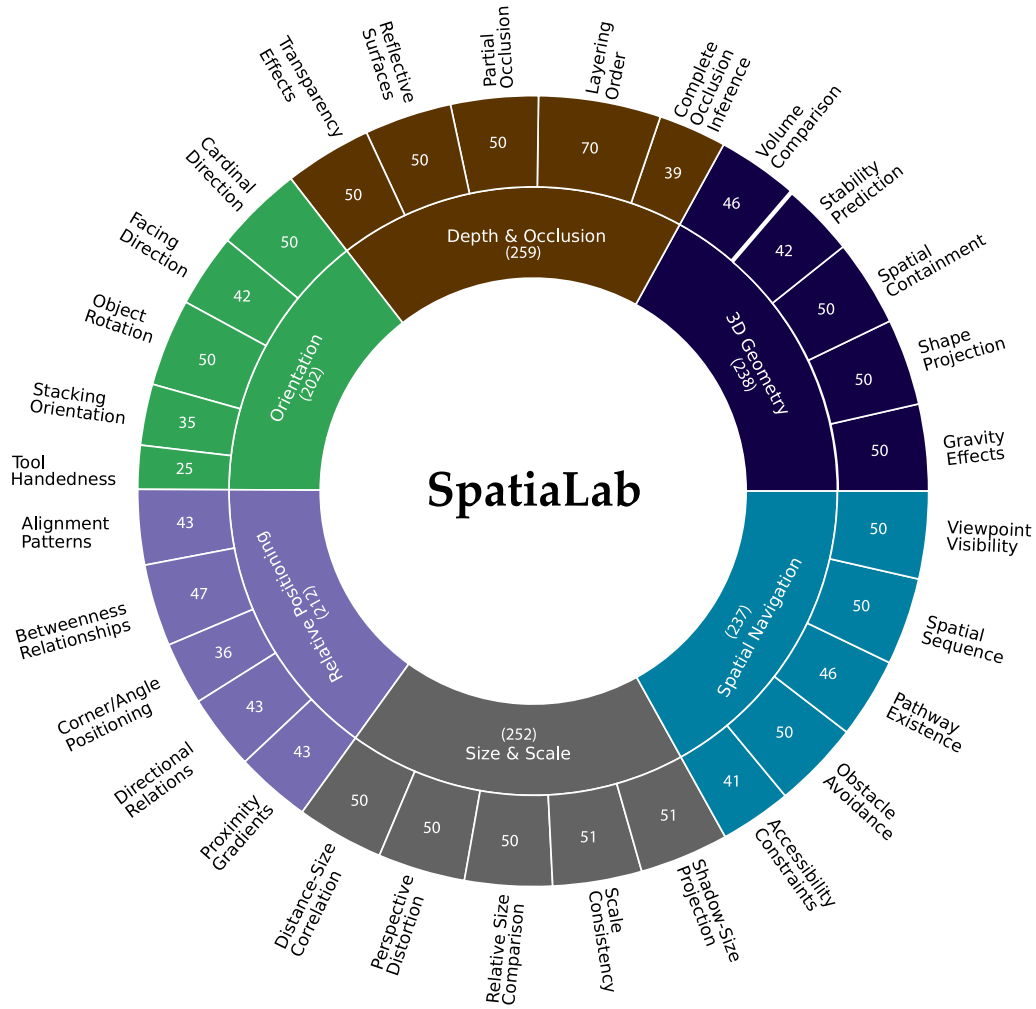


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each top-level category is decomposed into five subcategories, yielding thirty task types in total.

The following specifies, per category, the reasoning primitive targeted by each subcategory and its downstream application:

B.1 RELATIVE POSITIONING

Directional Relations. Directional relations specify the position of one object relative to another along axes such as left, right, above, or below. Resolving these relations enables a model to interpret spatial layouts in the manner of human scene descriptions; for example, localizing the object immediately to the left of an appliance supports scene understanding and robot navigation.

Proximity Gradients. Proximity reasoning requires ordering objects by their distance from a designated reference point, as in “Which object is nearest to the doorway?” A model that estimates such graded proximity supports navigation and assistive technologies.

1242 **Alignment Patterns.** Alignment reasoning determines whether objects follow a common orientation
1243 or ordering, such as vertical stacking or collinear arrangement. Detecting alignment supports
1244 interpretation of structured environments such as shelves, tables, and warehouses, and is critical for
1245 robotic sorting and automated inspection.

1246 **Betweenness Relationships.** Betweenness reasoning identifies the object that occupies the interval be-
1247 tween two references; determining what lies between the lamp and the armchair, for instance, requires
1248 reasoning over spatial continuity. This supports navigation, retrieval, and situational awareness.

1249 **Corner/Angle Positioning.** Corners and angled positions serve as reference anchors in spatial
1250 reasoning. Queries such as “What is positioned in the back-right corner?” require localizing objects
1251 within constrained regions, which is central to robotic mapping and placement.

1252

1253

1254 **B.2 DEPTH & OCCLUSION**

1255

1256 **Layering Order.** Layering reasoning requires ordering objects by front-to-back depth; identifying
1257 the frontmost item on a table, for instance, exercises depth estimation. This is particularly relevant to
1258 AR/VR applications, in which visual stacking determines user experience.

1259 **Partial Occlusion.** Partial occlusion arises when one object obscures part of another; recognizing
1260 the hidden item behind a coffee cup, for instance, requires reconstructing missing information. This
1261 improves robustness for perception systems operating in cluttered environments.

1262 **Complete Occlusion Inference.** When an object is fully occluded, indirect cues such as shadows or
1263 surface deformation reveal its presence; inferring what lies under a blanket requires reasoning from
1264 such indirect evidence, analogous to human inference under visual uncertainty.

1265 **Transparency Effects.** Transparent media, such as glass or water, admit partial visibility of the
1266 objects behind them; determining what is visible through a fish tank requires combining occlusion
1267 reasoning with transparency perception. This is central to simulation, graphics, and embodied AI.

1268 **Reflective Surfaces.** Reflective reasoning requires interpreting mirrors, specular metals, or glass
1269 reflections; identifying the reflected content of a mirror entails indirect spatial mapping. This is
1270 essential for safety in autonomous vehicles and situational awareness in robotics.

1271

1272

1273 **B.3 ORIENTATION**

1274

1275 **Cardinal Direction.** Cardinal reasoning concerns compass directions (north, south, east, west);
1276 determining the facing direction of a front door requires integrating global orientation cues, which is
1277 critical for navigation and geospatial alignment.

1278 **Object Rotation.** Rotation reasoning estimates the angular displacement of an object about an axis;
1279 estimating the rotation of a chair, for instance, is a prerequisite for accurate 3D modeling and supports
1280 correct object alignment for robotic manipulation.

1281 **Facing Direction.** Facing-direction reasoning specifies the orientation of an object’s front surface;
1282 determining whether a monitor faces north requires mapping its surface normal to a global reference
1283 frame, which is relevant to surveillance and robotics.

1284 **Stacking Orientation.** Stacking reasoning determines whether objects are arranged vertically,
1285 horizontally, or in mixed orientation; recognizing such configurations is a prerequisite for stability in
1286 warehouse and household settings and for safe robotic stacking.

1287 **Tool Handedness.** Certain tools, such as scissors or gloves, are designed for left- or right-handed use;
1288 inferring handedness requires fine-grained shape and orientation perception, which benefits assistive
1289 robotics in human-centered environments.

1290

1291

1292 **B.4 SIZE & SCALE**

1293

1294 **Relative Size Comparison.** Relative-size reasoning determines whether one object is larger, taller,
1295 or smaller than another; determining whether a vase is taller than a statue requires proportional
reasoning, which is central to packing, fitting, and recognition tasks.

1296 **Perspective Distortion.** Perspective projection can cause distant objects to subtend a smaller image
1297 than nearer objects independently of true size; recognizing such effects, such as a distant car appearing
1298 smaller than a nearby bicycle, requires compensating for depth cues and improves robustness to
1299 camera viewpoint.

1300 **Distance-Size Correlation.** Here size estimation is conditioned on distance estimation; a nearby
1301 bicycle subtending a larger image than a distant car must not corrupt the inferred true size. This
1302 supports navigation, AR/VR, and visual grounding.

1303 **Scale Consistency.** Scale reasoning verifies whether object sizes are mutually consistent within
1304 a scene; a mouse larger than a chair constitutes an inconsistency that a model must detect. This
1305 improves realism in simulated environments.

1306 **Shadow-Size Projection.** Shadows provide indirect cues to object size and light-source position;
1307 estimating the required height of a light source from shadow length exercises geometric reasoning,
1308 which is useful in graphics, architecture, and physics-based modeling.

1310

1311

1312 B.5 SPATIAL NAVIGATION

1313

1314 **Pathway Existence.** Pathway reasoning determines whether a traversable route exists between two
1315 points; determining whether the desk is reachable without displacing chairs tests navigability, which
1316 underpins indoor robot planning.

1317 **Obstacle Avoidance.** Obstacle avoidance requires computing collision-minimizing routes; identifying
1318 the clearest path from door to window maps directly to robotic and human navigation and is essential
1319 for autonomous driving and indoor mobility aids.

1320 **Viewpoint Visibility.** Visibility from a specified viewpoint requires determining which objects are
1321 occluded or visible; determining what is obscured when seated on the couch requires simulating the
1322 corresponding perspective, which improves AR/VR realism and robotic vision.

1323 **Spatial Sequence.** Sequence reasoning determines the order in which objects are encountered along
1324 a path; a traversal from the kitchen to the balcony may pass the dining table first. Such reasoning
1325 models human navigational memory.

1326 **Accessibility Constraints.** Accessibility reasoning identifies objects that are blocked from access by
1327 obstacles; a cabinet may be unreachable because of an intervening table. This is central to assistive
1328 AI and ergonomic design.

1329

1330

1331 B.6 3D GEOMETRY

1332

1333 **Volume Comparison.** Volume reasoning determines which object occupies more three-dimensional
1334 space; determining whether a box exceeds a sphere in volume supports packing and containment
1335 tasks and is relevant to logistics and simulation.

1336 **Stability Prediction.** Stability reasoning predicts whether adding or displacing an object will topple
1337 a structure; determining whether placing a book collapses a pile models human physical intuition and
1338 supports safety in robotics and construction.

1339 **Shape Projection.** Projection reasoning determines the 2D shape cast by an object under a given
1340 rotation or illumination; a sphere always projects a circle, whereas a cube may project a square or
1341 hexagon. This exercises geometric understanding.

1342 **Spatial Containment.** Containment reasoning determines whether one object can fit inside another;
1343 determining whether a ball fits inside a box supports packing and design and aligns with intuitive
1344 geometry.

1345 **Gravity Effects.** Gravity reasoning predicts motion outcomes under physical forces; a rolling ball
1346 may arrest under a table due to friction and barriers. This improves real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We recruited annotators aged 24 to 30 years, all of whom held engineering degrees. Annotators participated voluntarily and worked independently without external incentives. Prior to annotation, we administered structured training sessions to standardize task requirements and enforce consistency across annotators, and we maintained ethical and quality-control standards throughout. This protocol ensured that the resulting annotations were both reliable and consistent with the dataset design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Prior to annotation, we designed a structured training program to enforce consistency and reliability across the full task taxonomy. We first introduced annotators to the six top-level categories (*Relative Positioning, Depth & Occlusion, Orientation, Size & Scale, Spatial Navigation, and 3D Geometry*) and their thirty subcategories. For each subcategory, we supplied curated positive examples illustrating the target spatial relation (e.g., directional relations, occlusion inference, stability prediction) together with counter-examples that delineate common failure modes. Beyond the taxonomy, annotators were trained to condition question design on six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, so that questions explicitly exploited visual diversity. We demonstrated how these factors interact in real-world settings, such as occlusion inference under low-contrast lighting or scale-consistency estimation under perspective distortion.

We further introduced the two evaluation protocols, multiple-choice and open-ended. For the multiple-choice protocol, annotators were trained to construct precise distractors that test fine-grained discrimination; for the open-ended protocol, they were trained to formulate prompts that elicit compositional reasoning. To operationalize the training, we conducted calibration sessions in which annotators generated both question types for sample images and iteratively refined their designs under feedback. To situate task design within the broader literature, we reviewed prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) with annotators and analyzed their limitations. These resources predominantly emphasize simplified relations, synthetic or staged scenes, and a narrow reasoning-type set, while omitting natural complexity such as shadows, reflections, and gravity-sensitive stability. This gap analysis motivated the requirement for richer, more ambiguous, and depth-layered spatial contexts.

Finally, we provided detailed demonstrations of the annotation protocol, specifying how to select images with sufficient spatial complexity, how to map questions unambiguously to subcategories, and how to avoid trivial formulations. Annotators were instructed to prioritize tasks requiring multi-step reasoning, such as depth layering, relational chaining, and cross-view scale inference, over binary labels. Through this systematic training and iterative feedback, the final dataset balances taxonomic coverage against real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

The prompts supplied to the answer model are shown in Figure 6. The decoding configuration fixes temperature to 0 and `max_tokens` to 10, with all remaining generation parameters at their default values.

D.2 OPEN-ENDED EVALUATION

The prompts supplied to both the answer model and the judge model are shown in Figure 6. The decoding configuration fixes temperature to 0.7 and `max_tokens` to 1200, with all remaining parameters at their default values.



Figure 6: Prompts employed under the evaluation protocols.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To validate *Gemini-2.5-Flash* as an automatic judge, we constructed a class-balanced set of 240 samples spanning all categories. Each sample comprised the full evaluation context (question, ground truth, and model output) and was scored independently by three human annotators, who assigned binary labels (1 for correct and 0 for incorrect). We compared the judge’s decisions against the human annotations using a set of complementary agreement metrics, computed both overall and per category. Specifically, we report **Cohen’s Kappa**, **Accuracy against the Majority Vote**, and **Fleiss’ Kappa**, defined below.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies pairwise agreement between two raters with a correction for chance-level agreement. For a binary task it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o denotes the observed agreement and p_e denotes the chance-expected agreement: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c ; $\kappa = 1$ denotes perfect agreement, $\kappa = 0$ denotes chance-level agreement, and $\kappa < 0$ denotes systematic disagreement. We computed Cohen’s kappa between the judge and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}) and between the judge and the annotator majority vote (κ_{majority}).

Accuracy with Majority Vote. We additionally quantified the raw agreement between the judge and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} denotes the judge’s decision for sample i and y_i^{maj} denotes the majority vote across annotators. In contrast to Cohen’s kappa, accuracy does not correct for chance agreement and reports only the proportion of identical labels.

Fleiss’ Kappa. To quantify inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to more than two raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e =$

$\sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then given by $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Larger values denote stronger inter-annotator reliability beyond chance.

Table 5: Agreement metrics between Gemini-2.5-Flash as an automatic judge and three human annotators, computed per category. The table reports Cohen’s kappa against the majority vote and against each annotator, raw accuracy against the majority, and Fleiss’ kappa among the annotators. Bold values denote the highest agreement within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 shows that the LLM judge (Gemini-2.5-Flash) attains strong overall agreement with the human annotators, with a Cohen’s kappa of 0.738 against the majority vote and in the range 0.681–0.795 against individual annotators. Overall raw accuracy against the majority is 0.880, and the Fleiss’ kappa among human annotators is 0.774, indicating substantial inter-rater consistency. Certain categories, such as 3D Geometry, attain perfect agreement ($\kappa = 1.000$, accuracy = 1.000). Depth & Occlusion likewise attains high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) attain moderate agreement that nonetheless remains within acceptable reliability bounds.

These results indicate that Gemini-2.5-Flash scores correctness comparably to human evaluators, with Cohen’s kappa predominantly in the substantial range (0.61–0.80) and several categories reaching almost-perfect agreement (> 0.81). The Fleiss’ kappa of 0.774 confirms that the human annotators are themselves highly consistent, which validates the reliability of the reference labels. The judge attains perfect or near-perfect agreement on well-defined categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), evidencing accurate pattern recognition. The moderate agreement on Orientation and Size & Scale is attributable to task complexity and intrinsic subjectivity rather than to a deficiency of the judge. In aggregate, the LLM-as-judge protocol exhibits robust reliability and constitutes a practical automated scoring instrument that closely tracks human evaluation.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) attain some of the strongest results on the benchmark. The maximum is **GPT-4o-mini** at 85.71% on *Stacking Orientation*, the highest single-subtask accuracy overall. **Gemini-2.0-Flash** performs well on occlusion and projection, attaining 64.0% on *Partial Occlusion* and 64.71% on *Shadow-Size Projection*. Likewise, **GPT-5-mini** attains 68.0% on *Tool Handedness* and 55.71% on *Layering Order*. Deficits persist, as **Claude 3.5 Haiku** reaches only 32.0% on *Relative Size Comparison*. On average, closed-source systems exceed open-source systems by 15–25 percentage points across several categories. However, the within-model spread remains large, frequently exceeding 50 percentage points. We therefore conclude that closed-source training pipelines raise the achievable ceiling but do not confer uniform spatial competence.

Open-source Large Models. Scaling to large open-source models (Table 7) produces clear improvements alongside severe fragilities. **InternVL3.5-72B** attains 80.56% on *Corner/Angle Positioning*, among the highest open-source results, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** attain 66.67% on *Complete Occlusion Inference*. Deficits persist, however: **llama-3.2-11b-vision-instruct** attains only 2.0% on *Object Rotation*. Gains are also evident on *Stacking Orientation*,

Table 6: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

where **llama-3.2-90b-vision-instruct** attains 71.43%, but at the cost of accuracy on other categories. The within-model spread frequently exceeds 70 percentage points, indicating brittle specialization, and gravity reasoning remains modest at 56.0%. We find that scaling raises subtask peaks but does not resolve the underlying structural weaknesses; parameter count amplifies selective strengths while leaving foundational representational gaps intact.

Open-source Small Models. Small open-source models (Table 8) exhibit sharp contrasts in subtask accuracy. For example, **qwen-2.5-vl-7b-instruct** attains 74.29% on *Stacking Orientation*, whereas **InternVL3.5-1B** degrades to 12.0% on *Cardinal Direction*. Accuracy on *Pathway Existence* remains modest, with the best models tied at 34.78%, and gravity-related reasoning is comparably weak, at 50.0% for both **InternVL3.5-4B** and **Gemma-3-4B-it**. The within-model spread frequently exceeds 60 percentage points, indicating brittle generalization. These patterns indicate that small models are strongly cue-dependent, performing well where visual templates are salient but failing on tasks requiring deeper spatial abstraction, consistent with uneven representation learning across

Table 8: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

subcategories. In aggregate, small models provide localized competence but lack robustness across the full spectrum of spatial reasoning.

Reasoning Models. Reasoning-augmented models (Table 9) confer notable benefits on selected tasks but remain inconsistent in aggregate. **o4-mini** attains 65.12% on *Alignment Patterns* and 56.0% on *Gravity Effects*, exceeding most open baselines. **Gemini-2.0-Flash-Thinking** attains 62.75% on *Shadow-Size Projection* and 52.17% on *Volume Comparison*, evidencing stronger geometric reasoning. The maximum accuracies are attained by **Gemini-2.5-Flash-Thinking**, at 71.79% on *Complete Occlusion Inference* and 70.21% on *Betweenness Relationships*; yet the same family degrades elsewhere, to 34.0% on *Cardinal Direction* and 34.0% on *Shape Projection*. This variance indicates that reasoning mechanisms deepen accuracy on selected subcategories without broadening coverage, and within-model spreads frequently exceed 35 percentage points. We conclude that reasoning augmentation adds localized capacity while leaving structural gaps in spatial understanding: these models demonstrate promise but not generalizable competence.

Spatial Reasoning Models. Spatially specialized models (Table 10) yield targeted improvements but lack generality. **SpaceQwen2.5-VL-3B-Instruct** attains 48.0% on *Distance-Size Correlation* and 49.02% on *Scale Consistency*, clearly exceeding many open-source baselines on size-scale reasoning, and it leads on navigation, at 53.66% on *Accessibility Constraints* and 44.0% on *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** attain 50.0% on *Stability Prediction*, consistent with domain-specific reasoning. However, accuracy degrades sharply on other tasks, to only 24.0% on *Spatial Containment*, and these models rarely exceed 55.0% even on their strongest subcategories, with subcategory spreads frequently exceeding 30 percentage points. We find that specialization induces narrow peaks without holistic competence: domain targeting enhances selected capabilities but does not integrate them into a robust, general framework.

Overall Discussion. Across all categories (Table 8, Table 7, Table 6, Table 10, Table 9), the results exhibit consistent fragmentation. Small open-source models peak at 74.29% on *Stacking Orientation* but degrade to 12.0% on *Cardinal Direction*. Larger models raise the ceiling, e.g., 80.56% on *Corner/Angle Positioning*, yet still fail catastrophically at 2.0% on *Object Rotation*. Closed-source systems attain the maximum accuracies, with **GPT-4o-mini** reaching 85.71% on *Stacking Orientation*, yet fall to 32.0% on *Relative Size Comparison*. Specialized models exhibit strengths on navigation (53.66%) and stability (50.0%) but remain capped below 55% overall, and reasoning models attain localized gains (71.79% on *Complete Occlusion Inference*) alongside residual weaknesses (34.0% elsewhere). Within-model spreads frequently exceed 50 percentage points, indicating brittle internal consistency. Subcategories requiring physical abstraction, namely *Gravity Effects*, *Stability Prediction*,

Table 9: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Table 10: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

and *Spatial Containment*, remain below 35–40% across model families. We attribute these failures to two root causes: the absence of stable internal encodings of orientation and physics, and over-reliance on surface-level correlations. The wide variance, spanning 85.71% maxima and 2.0% minima, is consistent with pattern matching rather than robust spatial inference. Scaling, proprietary training, and reasoning augmentation each confer benefits, but none yields general coverage; each instead

Table 11: Distribution of item counts by the number of models (0–5) that answered each item correctly, per category and subcategory in SPATIALAB-MCQ. A value of “0” denotes items that no model answered correctly, and “5” denotes items answered correctly by exactly five of the tested models.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

produces isolated peaks coexisting with pronounced blind spots. The aggregate picture is one of fragmented rather than integrated competence. We conclude that progress in spatial reasoning will require architectural modifications that embed consistent geometric grounding and physics-aware mechanisms. In summary, the results simultaneously exhibit substantial selective gains and the persistence of fundamental gaps in holistic spatial intelligence.

E.2 MODEL FAILURE ANALYSIS

The count distribution across categories and subcategories in Table 11 exposes several regularities in model behavior. 3D Geometry and Size & Scale exhibit comparatively balanced distributions, yet both retain a substantial fraction of items that no model answers correctly, indicating that even

basic geometric invariances are not reliably internalized. Depth & Occlusion exhibits a heavier mid-range tail (2–4 correct), consistent with partial but inconsistent reasoning over layered scenes. Orientation is weak overall, with especially low totals on Stacking Orientation and Tool Handedness, indicating that subtle directional and embodied affordances are poorly captured. Relative Positioning is heterogeneous: Proximity Gradients attains moderate success, whereas Corner/Angle Positioning yields many near-zero counts, reflecting brittleness in compositional reasoning over reference frames. Spatial Navigation is comparatively stronger, particularly on Obstacle Avoidance and Viewpoint Visibility, but retains substantial error on Pathway Existence, which requires relational chaining. In aggregate, no category attains uniformly high accuracy, and the grand total is approximately uniform across the 0–5 bins, indicating that success is strongly task-dependent rather than reflecting general spatial competence. Overall, models exhibit isolated strengths but lack systematic reliability across spatial dimensions.

These results implicate several root causes tied to representational and reasoning bottlenecks. The high error rate on Stacking Orientation and Tool Handedness indicates limited grounding in physics-based and embodied reasoning, consistent with training predominantly on static internet imagery rather than interactive or task-driven data. The difficulty on Corner/Angle Positioning and Pathway Existence reflects inadequate multi-step relational chaining and a corresponding weakness in recursive spatial logic. The inconsistency on Transparency Effects and Reflective Surfaces indicates the difficulty of visual phenomena that require non-local cues, such as secondary reflections or occluded geometry. By contrast, the comparatively strong performance on Obstacle Avoidance is consistent with shortcut learning from dataset biases, such as frequent co-occurrence of walkable paths, rather than genuine planning. In aggregate, these failures indicate that current training regimes over-weight correlation-driven vision–language alignment while under-representing embodied, counterfactual, and physics-based reasoning. Remedying these deficits will likely require richer training distributions (e.g., embodied simulation data and physics-augmented learning) together with architectural mechanisms that explicitly model reference frames and relational constraints. Absent such advances, models will continue to exhibit brittle competence that degrades under compositional or physically grounded tasks.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models attain the strongest overall accuracy under the open-ended protocol, yet their results remain non-uniform across subcategories. GPT-5-mini is the clear leader, attaining the maximum accuracy on **3D Geometry** and **Relative Positioning**, with peaks of 58.14% on Directional Relations and 55.81% on Alignment Patterns. In contrast, Gemini-2.5-Flash and Claude 3.5 Haiku attain very low accuracy on Proximity Gradients (16.28%) and Perspective Distortion (12.0%), respectively. This wide gap quantifies the fragility of even the strongest models when reasoning over subtle, continuous visual cues. We attribute these disparities to differing degrees of sophistication in the models’ spatial reasoning pipelines, with some architectures capturing relational and geometric structure more effectively than others. Tasks such as Proximity Gradients are particularly demanding, as they require analogical, graded estimation rather than discrete classification. That GPT-5-mini succeeds where others fail is consistent with scaling and proprietary training corpora conferring distinct advantages. Nevertheless, even the leader exhibits vulnerabilities, indicating a systemic property: closed-source dominance is relative rather than absolute, and these models remain short of robust, human-level spatial reasoning.

Open-source Large Models (Table 13). Large open-source models exhibit a more dispersed accuracy landscape, with different systems attaining peaks on isolated subcategories. llama-3.2-90b-vision-instruct attains 60.0% on Depth and Occlusion, whereas GLM-4.5V-106B-MoE leads on Spatial Containment at 38.0%. Despite these peaks, several models degrade sharply on Proximity Gradients, with qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct both attaining only 9.3%. This indicates that architectural scaling assists but does not guarantee coverage on difficult continuous-reasoning tasks. We attribute much of this variation to heterogeneous training data: models exposed to richer depth and geometry corpora perform better on occlusion and containment tasks, whereas others generalize poorly. The persistent weakness on Proximity Gradients indicates that these systems are not optimized for subtle spatial analogies and distance scaling. We further observe higher inter-

Table 12: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	30.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	24.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

model variance in this group than among closed-source models, consistent with the heterogeneity of open-source development. In aggregate, large open-source models exhibit encouraging peaks but inconsistent coverage, indicating that scale alone is insufficient absent targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models lag their larger counterparts, as expected, but exhibit noteworthy localized strengths. qwen-2.5-vl-3b-instruct attains 18.0% on Perspective Distortion and 26.0% on Relative Size Comparison, whereas Gemma-3-4B-it leads on Spatial Containment (24.0%) and Stability Prediction (21.43%). These peaks are offset by catastrophic failures: InternVL3.5-2B attains 0.0% on Betweenness Relationships, and InternVL3.5-1B attains only 2.0% on Shape Projection. We attribute this extreme variance to limited capacity and restricted training diversity, which preclude the learning of robust geometric and relational representations. Subcategories that demand precise orientation and hierarchy reasoning induce complete breakdowns,

Table 14: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

since small misinterpretations propagate to total failure. Nevertheless, the occasional advantage of small models over larger ones on narrow tasks indicates that even small-scale training can instill isolated competencies, evidencing the value of specialization while confirming the structural ceiling of compact architectures. In summary, small open-source models exhibit localized promise but remain highly brittle, with overall accuracy bounded by capacity and data constraints.

Reasoning Models (Table 15). Reasoning-oriented models exhibit a pronounced internal divide, with Gemini-2.5-flash-thinking substantially above its peers. It attains the maximum accuracy on nearly every subcategory, including 75.0% on Tool Handedness and 57.14% on Complete Occlusion Inference. By contrast, Kimi-VL-A3B-Thinking-2506 performs poorly throughout, attaining only 6.0% on Spatial Containment. We attribute this gap to differences in how the models integrate multimodal streams into a coherent reasoning process. Gemini-2.5-flash-thinking appears able to construct and query internal world models, enabling more effective inference of object interactions and physical properties, whereas weaker models rely on surface-level pattern matching that is inadequate for compositional or causal reasoning. The consistently high accuracy of Gemini-2.5 across categories indicates that explicit reasoning mechanisms confer measurable benefits for spatial understanding. However, its failure to lead on Shape Projection and Betweenness Relationships indicates that even advanced reasoning modules are not yet universally effective. Thus, while reasoning models are promising, the category remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, though explicitly designed for the task, attain the lowest overall accuracy. Their performance is highly fragmented: several models tie at 16.0% on Gravity Effects, 8.0% on Shape Projection, and 14.29% on Stability Prediction. SpaceQwen2.5-VL-3B-Instruct is marginally stronger, at 23.91% on Volume Comparison and 13.73% on Scale Consistency, but these values remain far below competitive thresholds. At the opposite extreme, Betweenness Relationships and Corner/Angle Positioning collapse entirely, at 0.0% and 2.78% respectively. These results indicate that the models’ limited capacity and narrowly focused training severely constrain their representational flexibility. We interpret the failures as evidence that narrow specialization, absent sufficient capacity or data breadth, cannot yield generalizable spatial reasoning; on geometric and positional tasks in particular, even slight representational errors propagate to total breakdown. Consequently, despite their explicit focus, spatial reasoning models are the least effective group, indicating that both scale and diversity are necessary for robust performance.

Overall Discussion. Across all model groups, closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16), the

Table 15: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

results exhibit a consistent pattern of fragmented competence and systemic weakness. Closed-source models attain the highest overall accuracy, with GPT-5-mini reaching 58.14% on Directional Relations and 55.81% on Alignment Patterns, yet degrade on Proximity Gradients. Large open-source models attain notable peaks, e.g., 60.0% on Depth and Occlusion, but fall to 9.3% on Proximity Gradients,

demonstrating that scale without specialization is insufficient. Small open-source models corroborate this, with rare localized strengths (e.g., 26.0% on Relative Size Comparison) and catastrophic minima (e.g., 0.0% on Betweenness Relationships). Reasoning models demonstrate that explicit reasoning mechanisms can substantially improve accuracy, with Gemini-2.5-flash-thinking reaching 75.0% on Tool Handedness, whereas peers such as Kimi-VL-A3B fall to single digits, exposing the fragility of less-integrated reasoning pipelines. Spatial reasoning models, despite their specialization, are the weakest group, frequently below 15% and failing entirely on Betweenness Relationships. In aggregate, the results implicate three root causes: (1) an absence of robust representations for continuous and analogical spatial cues, (2) structural limitations coupled to scale and training diversity, and (3) brittle reasoning pipelines that preclude cross-task generalization. We conclude that no current approach, whether proprietary scaling, open-source expansion, reasoning integration, or specialization, yields comprehensive coverage; the frequent within-model gaps exceeding 50 percentage points further confirm that progress is piecemeal. Future work should therefore prioritize unified geometric encodings, physics-informed reasoning, and cross-task transfer, advancing beyond isolated gains toward consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The count distribution in Table 17 localizes persistent weaknesses across categories, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry is broadly difficult, with Shape Projection (19 zero-correct items) and Stability Prediction (10 zero-correct items) exhibiting especially poor outcomes, consistent with unreliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, accumulating the highest zero-correct count (72), concentrated in Layering Order (20 zero) and Transparency Effects (16 zero), both of which require fine-grained handling of occlusion and non-local cues. Orientation exhibits analogous difficulty: Cardinal Direction attains mid-range success, whereas Stacking Orientation (11 zero) and Tool Handedness (5 zero, none solved by all models) indicate the brittleness of embodied directional reasoning. Relative Positioning is broadly inconsistent, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models and Corner/Angle Positioning low but comparatively balanced. Size and Scale is relatively stronger, with several subcategories (e.g., Scale Consistency at 6 full successes) indicating that some models internalize coarse perspective and scaling cues, though the zero-correct count remains high on Shadow-Size Projection (17 zero). Spatial Navigation is bimodal: Spatial Sequence and Obstacle Avoidance attain occasional full-model success, whereas Pathway Existence (16 zero) and Accessibility Constraints (10 zero) indicate persistent difficulty with multi-step relational chaining. In aggregate, the zero-correct bin dominates the distribution while only 74 items are solved by all five models, confirming that consistent spatial competence remains elusive.

These findings indicate that the root causes of failure derive less from dataset design and more from the representational and reasoning limitations of current models. First, the concentration of errors in Depth and Occlusion reflects reliance on superficial appearance features rather than structured scene representations: models attend to surface texture rather than reasoning over hidden layers, yielding systematic errors on Layering Order. Failures on Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretraining imagery typically lacks explicit left/right-handedness or stacking-physics annotation. The brittleness on Relative Positioning and Spatial Navigation indicates deficient multi-hop relational chaining; Pathway Existence, for instance, requires integrating local connectivity with global layout, which attention-based co-occurrence priors do not capture. By contrast, partial success on Size and Scale is consistent with exploitation of statistical regularities in perspective distortion and scale consistency, which nonetheless collapse under rarer cases such as shadows. Collectively, these patterns indicate that current models succeed when cues are salient and correlated with training priors but fail when the task requires counterfactual, compositional, or embodied inference. Remedying these deficits will require architectural mechanisms that explicitly model reference frames, occlusion layers, and physical dynamics, together with training data enriched by embodied simulation and counterfactual augmentation. Absent these, progress in spatial reasoning will remain fragmented, with models performing well on surface-level cues but degrading on structured, real-world spatial logic.

Table 17: Distribution of item counts by the number of models (0–5) that answered each item correctly, per category and subcategory in SPATIALAB-OPEN. A value of “0” denotes items that no model answered correctly, and “5” denotes items answered correctly by exactly five of the tested models.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This section provides a detailed analysis of the accuracy gap between the multiple-choice (MCQ) and open-ended protocols across the spatial reasoning categories of SPATIALAB, as reported in Table 18.

G.1 QUANTITATIVE SNAPSHOT

Aggregated over 25 models, the mean MCQ→open-ended accuracy gap is **23.0%** with a standard deviation of 5.5%. The dataset-wide per-subtask means are: 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation

Table 18: **Accuracy gap between the MCQ and open-ended protocols (%) ($\downarrow$) across spatial categories on SPATIALAB.** A lower value denotes comparable accuracy under the two protocols, whereas a higher value denotes greater disparity. The largest gap within each model group is bold and underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

(22.89), for an overall mean of 23.01. The overall gap ranges from a minimum of 11.93 to a maximum of 31.57, with a median of approximately 23.34. At the group level, spatial reasoning models exhibit the largest mean gap (27.03), whereas reasoning-oriented models exhibit the smallest (19.11); proprietary models exhibit a lower mean gap (21.09) than many large open-source models (24.51). The human reference exhibits an overall gap of 22.64. The largest gaps occur for InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), and the smallest for Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlation analysis identifies the spatial-navigation gap as the dominant predictor of the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) as strong secondary predictors.

G.2 PATTERNS AND OBSERVATIONS

Three broad patterns emerge. First, specialist spatial reasoning models exhibit the largest MCQ→open-ended gaps, consistent with optimization for classification-style supervision and reduced robustness to free-form generation; reasoning-focused models, by contrast, exhibit smaller gaps, consistent with the stabilizing effect of chain-of-thought-style training on generative outputs. Second, the spatial-navigation gap is the strongest predictor of the overall disparity: models that degrade on open-ended navigation reasoning drive the aggregate gap, with orientation and 3D geometry as significant secondary contributors and relative position and size/scale varying more idiosyncratically across models. Third, scale alone does not account for robustness: smaller models such as Llama-3.2-11B attain among the lowest gaps, whereas very large models such as InternVL3.5-72B attain the highest. Finally, several negative or near-zero gaps occur (e.g., -1.98 for Llama-3.2-11B on depth & occlusion and -3.18 for o4-mini on relative position), indicating that the open-ended protocol is, in these cases, the more reliable estimator, most plausibly because of misleading distractors.

G.3 HYPOTHESES ON ROOT CAUSES

We posit several interacting factors underlying the observed MCQ→open-ended gaps in SPATIALAB, each supported by the statistical evidence:

1. **MCQ structural advantage.** The MCQ format constrains the output space, admitting exploitation of surface cues and distractor elimination. Across models, the mean MCQ accuracy systematically exceeds the open-ended accuracy, with a mean gap of 23.0% ($\sigma = 5.5\%$). The negative gaps for Llama-3.2-11B on depth & occlusion (-1.98) and o4-mini on relative position (-3.18) show that open-ended generation can occasionally exceed MCQ when distractors are misleading, corroborating that the MCQ protocol can inflate or distort measured performance.

Recommendation: We recommend complementing MCQ with open-ended evaluation and subtask-level diagnostics, which recover model competence more accurately and suppress the overestimation induced by option-based cues.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models exhibit the largest mean gap (27.03%) despite being optimized for spatial tasks, consistent with training predominantly on categorical or synthetic selection objectives rather than free-form generation. For example, the 72B-parameter InternVL3.5-72B exhibits the largest overall gap (31.57%), concentrated in spatial navigation (36.68%) and orientation (34.44%), indicating reliance on classification-style supervision. The correlations between subtask and overall gaps (spatial navigation $r = 0.99$, orientation $r = 0.83$) indicate that these format-dependent failures concentrate in sequential and geometric reasoning tasks.

Recommendation: We recommend fine-tuning or instruction-tuning on open-ended spatial generation, which improves robustness to generative formats and reduces reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models (e.g., Gemini-2-Flash-Thinking, o4-mini) exhibit smaller mean gaps (approximately 19.11%) and lower subtask variance. Gemini-2-Flash-Thinking, for example, exhibits an overall gap of 13.0% and consistently modest subtask gaps (3D geometry 13.9%, relative positioning 21.03%), consistent with chain-of-thought or stepwise decoding yielding more stable open-ended responses. The reduced correlation between subtask-gap variance and overall gap within this group further indicates that instruction-tuned reasoning attenuates format sensitivity.

Recommendation: We recommend chain-of-thought or stepwise decoding together with instruction-tuning for open-ended spatial tasks, which improves generative consistency and reduces MCQ dependence.

4. **Spatial navigation stresses sequential grounding.** The spatial-navigation gap is the strongest predictor of the overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models with large spatial-navigation gaps, such as SpaceQwen2.5-VL-3B-Instruct (spatial-navigation gap = 35.87%), also exhibit large overall gaps (29.78%), indicating that multi-step referential reasoning is disproportionately affected under the open-ended protocol. This is consistent with MCQ options simplifying the reasoning chain and thereby masking weaknesses in sequential grounding and relational computation.

Recommendation: To characterize sequential reasoning more faithfully, we recommend dedicated open-ended spatial-navigation tasks and targeted multi-step reasoning supervision, which improve grounding and reduce MCQ overestimation.

5. **MCQ distractor artifacts.** The negative and near-zero gaps indicate that poorly calibrated distractors can suppress measured MCQ accuracy. o4-mini, for example, exhibits a negative relative-position gap (-3.18%) while maintaining positive gaps on other subtasks, indicating that distractor quality can artificially depress MCQ accuracy. Conversely, Qwen-VL2.5-32B-Instruct exhibits very large MCQ→open gaps (overall = 29.85%) across multiple subtasks, indicating that distractors can also inflate the perceived competence in selection-based tasks.

Recommendation: We recommend systematic auditing and refinement of MCQ distractors together with open-ended evaluation, which yields more accurate and less biased estimates of spatial reasoning competence.

G.4 PRACTICAL IMPLICATIONS

These findings establish the necessity of reporting both MCQ and open-ended results. Exclusive reliance on MCQ inflates estimates of practical spatial competence, with a mean disparity of approximately 23%. Subtask-level breakdowns are essential, since the spatial-navigation gap disproportionately drives the observed differences. Diagnostic evaluations that separate discriminative from generative reasoning more precisely isolate the error modes. Furthermore, instruction-tuning and decoding strategies that promote stepwise reasoning are promising for reducing MCQ dependence. Finally, systematic design and auditing of MCQ distractors are required to ensure fair cross-protocol comparison.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To improve spatial reasoning accuracy on the benchmark, we evaluated multiple complementary intervention strategies.

H.1 INHERENT REASONING MECHANISMS

We next quantify the contribution of built-in reasoning mechanisms by comparing standard models against their reasoning-enabled counterparts (Table 2, Table 3). Under the MCQ protocol, reasoning consistently increases accuracy, with a category-dependent magnitude. Gemini-2.5-Flash-Thinking, for example, exceeds its non-reasoning variant by +4.64 points overall (52.93% vs. 48.29%), with the largest gains on Relative Positioning (+13.1%) and Orientation (+8.6%). The same trend holds for o4-mini-medium, which attains the highest overall MCQ accuracy (53.21%), consistent with explicit reasoning scaffolds being especially effective in structured, discrete-answer settings.

Under the open-ended protocol the picture is more nuanced. Reasoning-equipped models again exceed their baselines, but the gains are non-uniform. Gemini-2.5-Flash-Thinking modestly exceeds Gemini-2.5-Flash (32.77% vs. 30.93%), while the largest gain is attained by o4-mini-medium (37.86%), which substantially exceeds all non-reasoning models. These results indicate that reasoning mechanisms attenuate the instability of free-form generation, particularly on complex categories such as 3D Geometry and Spatial Navigation. The gains are not universal, however: reasoning models occasionally regress, e.g., Gemini-2.5-Flash-Thinking degrades sharply on Size & Scale (21.74% vs. 26.59%), corroborating the earlier finding that the generative setting amplifies brittleness.

In aggregate, these comparisons establish that explicit reasoning modules substantially improve structured MCQ accuracy and can stabilize open-ended responses, but with a category-dependent benefit. The results indicate that reasoning mechanisms strengthen logical consistency and relational chaining yet do not resolve scale sensitivity, occlusion, or free-form linguistic grounding.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To determine whether structured reasoning improves spatial accuracy, we further evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 subcategories in SPATIALAB, we sampled 5 items (random seed fixed to 42) to construct a class-balanced test set covering all major spatial phenomena. This design ensures balanced coverage while bounding the evaluation cost. Representative CoT prompts, illustrating the step-by-step reasoning scaffolds supplied to the models, are shown in Figure 7. This protocol isolates the contribution of CoT reasoning from model scale and training.

The analysis (Table 19) shows that orientation is the only category on which CoT prompting reliably increases accuracy, whereas most other categories exhibit stagnation or decline. This indicates that CoT assists when a task is reducible to the logical alignment of directions but fails when it requires richer perceptual priors (e.g., depth, scale, relative positioning). We attribute this to the absence of internalized 3D-perception and geospatial knowledge in the models; they are not intrinsically “aware” of these concepts in the human sense. Consequently, applying CoT does not correct misconceptions but reinforces them, propagating and amplifying errors across reasoning steps.

This result establishes that the benchmark is non-trivial: it is not solved by appending multi-step reasoning to existing models. In contrast to textual reasoning tasks, on which CoT reliably improves

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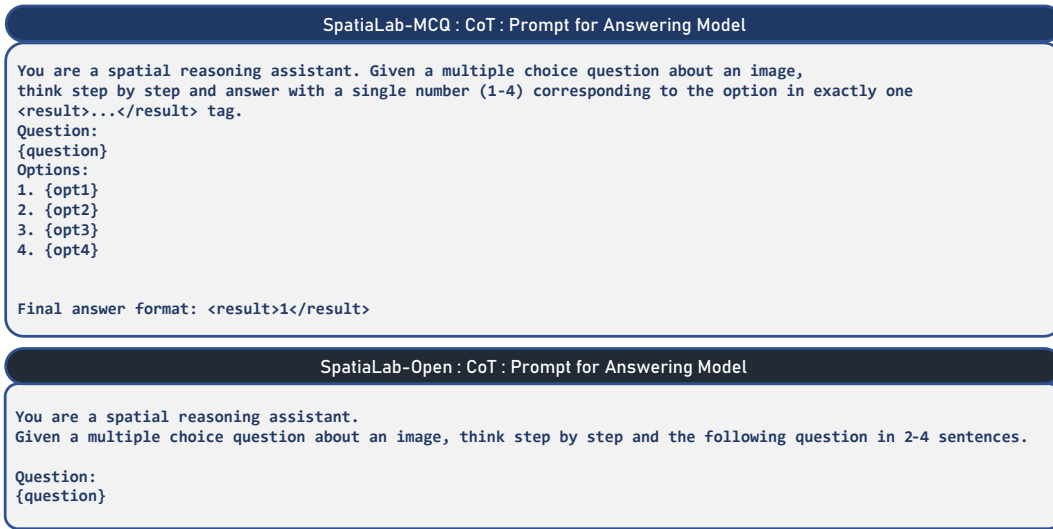


Figure 7: Chain-of-Thought (CoT) prompts.

Table 19: Per-category accuracy under CoT prompting for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); the Gain column reports the with-CoT minus without-CoT difference.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

outcomes, this benchmark exposes deeper architectural limitations in perceptual and spatial grounding. On categories such as depth and size, an incorrect prior implies that each additional reasoning step moves the prediction further from the correct answer. The sharp accuracy reductions, most pronounced for Gemini under the MCQ protocol, quantify how fragile reasoning becomes when it is not supported by robust perceptual knowledge.

The task design is therefore a strength: it moves beyond surface-level reasoning to test whether models possess genuine geospatial and 3D relational understanding. Solving it requires more than logical chaining; it requires representations and training that capture the core structure of perception and spatial cognition. In this respect, the benchmark stresses models at a fundamental level and localizes an important gap in current architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We extend the CoT protocol with a self-reflection stage, in which the model is prompted to review and revise its initial reasoning before emitting a final answer (see Figure 8). The evaluation protocol is otherwise identical to the standard CoT setting, using 5 class-balanced samples per subcategory (seed 42). This design isolates the effect of self-reflection while keeping the comparison directly aligned with the baseline CoT condition.

Augmenting CoT prompting with self-reflection yields mixed and strongly model-dependent outcomes. For InternVL-3-78b, overall accuracy declines slightly (-2.66% MCQ, -8.67% open-ended), with sharp reductions on depth, relative positioning, and navigation, consistent with reflection reinforcing rather than correcting errors. Localized gains occur, such as $+12\%$ on Size & Scale (MCQ), but they are inconsistent and offset by large degradations (-28% on relative positioning, open-ended). By contrast, Gemini-2.5-Flash exhibits large MCQ gains ($+20.66\%$), with the largest

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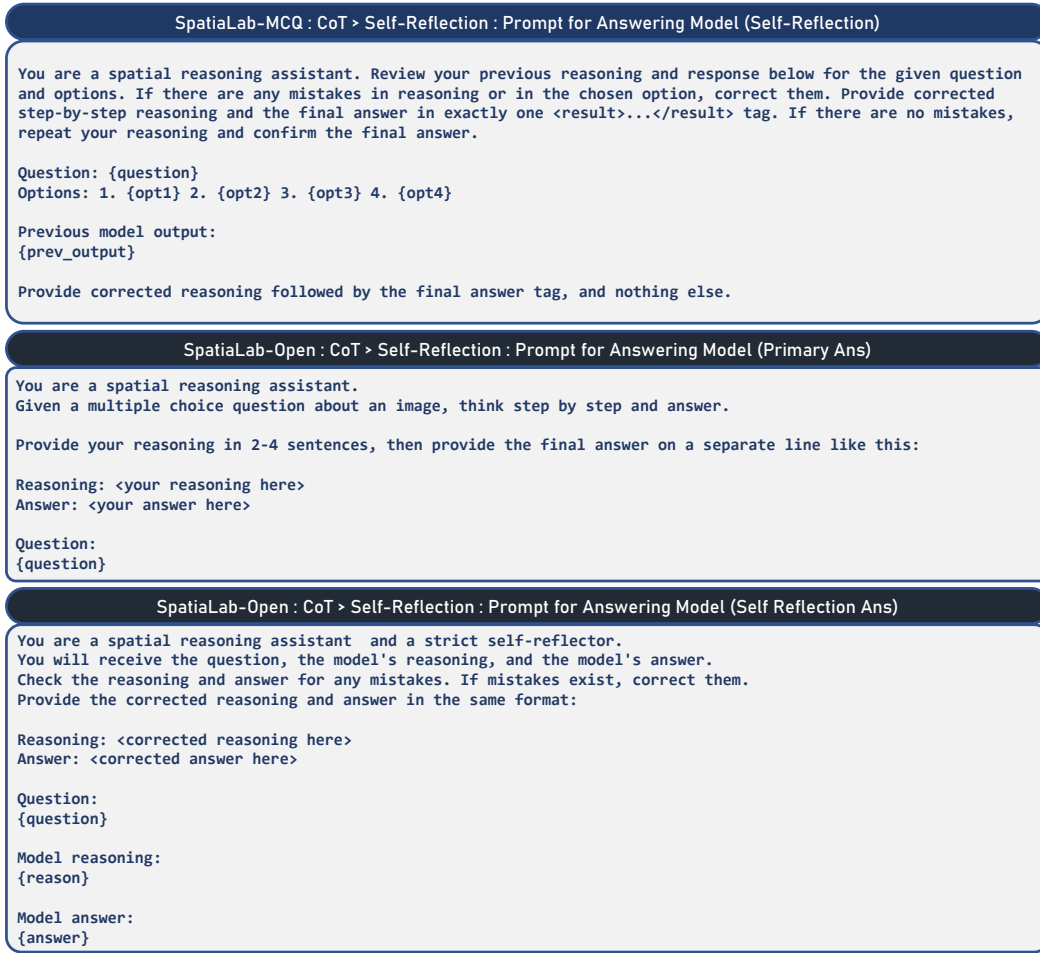


Figure 8: Chain-of-Thought (CoT) with self-reflection prompts.

Table 20: Per-category accuracy under CoT vs. CoT with self-reflection (CoT-SR) for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); the Gain column reports the CoT-SR minus CoT difference.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

improvements on 3D Geometry (+32%) and Depth & Occlusion (+24%), while its open-ended improvement remains negligible (+0.67%). These results expose a key asymmetry: reflection helps under the multiple-choice protocol, where the model can prune unlikely options, but does not transfer to free-form generation, where uncertainty propagates into ungrounded rationales. We further find that the categories most improved by reflection are those in which geometric consistency is central (orientation, depth, size), whereas tasks requiring relational integration across multiple objects (relative positioning, navigation) degrade. In aggregate, reflection functions as a stabilizer under structured choice but does not reliably improve naturalistic reasoning.

This inconsistency mirrors the plain-CoT result: reasoning augmentations succeed only when the model possesses stable perceptual anchors on which to ground its steps. Self-reflection introduces an explicit verification pass, but its efficacy is bounded by the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies rather than repairs misconceptions. This accounts for Gemini, with stronger perceptual grounding, benefiting under the MCQ protocol, while InternVL-3-78b degrades more frequently. The broader root cause is a pretraining mismatch: current VLMs import reasoning templates from language modeling but lack the embedded 3D and geospatial representations that humans employ. Absent these anchors, reflection cannot perform true error correction and instead functions as a linguistic filter that refines reasoning syntax without improving perceptual grounding. Consequently, reflective reasoning loops provide modest utility under discrete-answer formats but collapse under open-ended demands, where factual verification is absent.

These findings confirm that the benchmark exposes a fundamental gap in current architectures: surface-level reasoning layers alone cannot compensate for absent spatial cognition, and meaningful progress will require deeper integration of geometric and perceptual structure. Moreover, because VLMs are typically trained on web data, concepts that are well-represented online correspond to categories on which models perform better, whereas poorly represented spatial and 3D concepts suffer, reinforcing these disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using parameter-efficient adaptation with LoRA (Hu et al., 2021). To operate within the available GPU budget, we loaded the model in 4-bit precision (load_in_4bit=True) and inserted LoRA adapters into the attention and feed-forward projection layers. The LoRA hyperparameters, comprising rank, scaling factor, dropout, and target modules, are specified in Table 21. Gradient checkpointing was enabled to reduce peak memory, and a fixed random seed enforced reproducibility.

For tokenization, we applied the Qwen-2.5 chat template to maintain consistent formatting between training and inference. Each training example was serialized into a single text sequence, and dynamic padding was applied via DataCollatorForSeq2Seq. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). The trainer and optimization hyperparameters, comprising learning rate, optimizer, weight decay, and scheduler, are specified in Table 22. The training loop comprised four passes, each of a single epoch, yielding four checkpoints.

In aggregate, this configuration enables efficient fine-tuning of a large vision-language model within a constrained GPU budget while retaining high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trajectories in Figure 4a reveal a marked divergence between the MCQ and open-ended protocols over training epochs. MCQ accuracy starts from a lower baseline of 25.63% and increases monotonically, converging near 36.59% by the fourth epoch. This monotone improvement indicates that the model benefits strongly from additional supervised fine-tuning, particularly on tasks that constrain answers to a bounded candidate set. Open-ended accuracy, by contrast, starts higher at 34.40% but declines sharply by the second epoch, reaching a minimum of 12.62%, and only partially

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

recovers in later epochs to 35.48% at termination. This instability indicates a sensitivity of open-ended generation that is absent under the MCQ protocol, where the multiple-choice structure supplies a more stable optimization signal. We interpret the open-ended fluctuations as evidence that fine-tuning overfits to MCQ-style reasoning at the expense of free-form linguistic generalization. The dashed baselines in the plot further show that the MCQ protocol improves beyond its initial value while the open-ended protocol remains approximately stationary. In aggregate, SFT preferentially benefits MCQ tasks in both stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure with a before-and-after breakdown across the six spatial reasoning categories. Under the MCQ protocol, accuracy improves in every category, with gains ranging from 1.59% on Relative Positioning to 17.11% on Size and Scale, and a grand-total gain of 10.97%, confirming that fine-tuning improves structured-answer accuracy. Under the open-ended protocol, the results are heterogeneous, comprising both positive and negative changes: 3D Geometry and Spatial Navigation exhibit small positive gains (3.50% and 2.82%, respectively), whereas Size and Scale decreases by -3.95% and Relative Positioning by -0.79% , indicating that fine-tuning can degrade certain reasoning modes in free-form contexts. The overall open-ended gain is only 1.07%, negligible relative to the MCQ improvement. These results establish that SFT efficacy is strongly conditioned on the evaluation protocol, and the per-category breakdown localizes where training helps, where it fails, and how the benefit differs between constrained and unconstrained outputs.

A finer inspection reveals pronounced disparities across categories and protocols. Under MCQ, Size and Scale and Spatial Navigation yield the largest gains, indicating that multiple-choice supervision is especially effective for scale discrimination and navigational reasoning. Under the open-ended protocol, however, Size and Scale declines, demonstrating that the same learned representations do not transfer to generation. Similarly, Relative Positioning improves marginally under MCQ and degrades under open-ended, indicating a consistent relational-reasoning weakness independent of protocol. Conversely, Orientation exhibits large MCQ gains ($+13.33\%$) and small open-ended gains, indicating that some categories generalize more robustly. We attribute the protocol divergence to the structural scaffolding of MCQ, which reduces the demand on the model to produce precise linguistic formulations, whereas the open-ended protocol requires the simultaneous mastery of reasoning and linguistic fluency, which amplifies the underlying weaknesses. The trends indicate that fine-tuning strongly improves discriminative pattern recognition while introducing volatility and even degradation under free-form generation, so that interventions optimized for MCQ do not transfer seamlessly to unconstrained reasoning.

These findings have two implications. First, SFT as applied here confers strong benefits on structured tasks but limited or negative transfer to open-ended reasoning. This limitation is not solely a fine-tuning artifact but also reflects the pretraining regime of contemporary VLMs. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning while providing little explicit supervision of compositional spatial structure, occlusion layers, or geometric transformation. Consequently, models excel at surface-level associations but fail to acquire robust

representations of depth, orientation, or scale. The observed instability under the open-ended protocol indicates that the learned priors are fragile: absent multiple-choice scaffolding, they revert to shallow pattern matching rather than grounded inference. The categories with the largest declines, Size and Scale and Relative Positioning, indicate missing inductive biases in pretraining, biases that humans acquire through embodied interaction with 3D environments. A further contributing factor is reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions and thereby impede generalization to real-world clutter, noise, and complexity. These findings motivate reformulating both pretraining and fine-tuning for spatial reasoning around datasets and objectives that reflect naturalistic visual-spatial challenges. Absent such targeted reforms, current VLMs risk an algorithmic constraint whereby they appear competent under constrained evaluation yet remain brittle and misaligned with the embodied, flexible reasoning that humans apply to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To quantify the stability of supervised fine-tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) with the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, exhibits a clear divergence in the learning dynamics between the multiple-choice (MCQ) and open-ended protocols.

Divergent Learning Trajectories. As reported in Table 23 and Figure 9, accuracy on SPATIALAB-MCQ improves monotonically, from a baseline of 25.63% to a mean of 35.74% at Epoch 4. This indicates that the model adapts to the discriminative structure of the task, mapping visual features to the constrained output set with increasing reliability.

In sharp contrast, SPATIALAB-OPEN exhibits a pronounced *U-shaped* trajectory across all seeds. Rather than monotone improvement, accuracy collapses in the early epochs (from 34.40% to a mean of 19.26% at Epoch 1) and, while recovering in later epochs (converging to $\approx 34.82\%$ at Epoch 4), yields a negligible final gain over the zero-shot baseline ($+0.42\%$).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). MCQ accuracy improves monotonically, whereas open-ended accuracy exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability under generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

Representational Brittleness and Alignment Tax. We attribute this phenomenon to the fragility of spatial representations in current VLMs under fine-tuning. The initial collapse in open-ended accuracy is consistent with *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation: as the model optimizes for the instruction-following format of the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible generative spatial grounding required for open-ended description.

The subsequent recovery phase corresponds to the model re-learning to articulate spatial concepts within the new distribution, yet it does not significantly exceed the initial baseline. This dynamic supports a central finding: SFT is effective for teaching a model to *select* correct spatial answers but

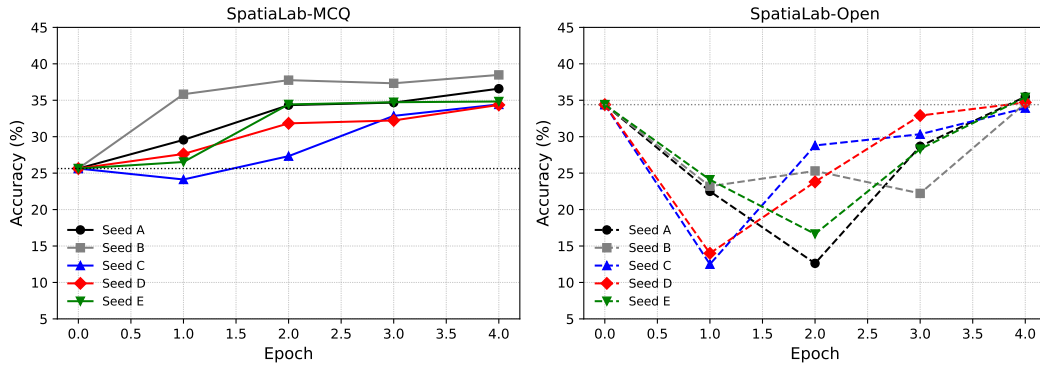


Figure 9: Analysis of SFT training dynamics.

does not fundamentally improve the underlying *generative spatial representation* required for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To determine whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we quantified the transferability of the fine-tuned model to three external spatial reasoning datasets: OMNISPIATIAL, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol specified in Appendix H.4.1, and evaluated it on each external benchmark in the zero-shot setting.

Table 24: Zero-shot transfer accuracy of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model acquires generalizable spatial representations rather than overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As reported in Table 24, fine-tuning on SPATIALAB yields substantial and consistent accuracy gains across all evaluated domains, ranging from +5.24% to +11.70%. The largest gain occurs on MIND THE GAP (+11.70%), consistent with the benchmark’s emphasis on occlusion and spatial continuity addressing the reasoning gaps targeted by that dataset. The gains on OMNISPIATIAL (+7.05%) and SPACE (+5.24%) similarly confirm that the geometric and relational concepts encoded in our taxonomy align with the broader definitions of spatial intelligence in the literature. These results validate the quality of the SPATIALAB data. Although the primary purpose of this work is to provide a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely induce overfitting to a specific question template; rather, the diversity of the 30 subcategories requires the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), for structured spatial reasoning over images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each assigned to a dedicated agent, which enables focused processing of objects, attributes, relations, symmetries, and transformations. This modular design improves accuracy and interpretability and supports multi-step reasoning, propagating low-level perceptual cues through higher-order relational and transformational inference.

The architecture, the per-agent implementation, and the coordination protocol are specified in the following sections; the complete prompt set is provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. The base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, enumerating object types (geometric shapes, natural elements, and artificial icons) together with attributes such as size, orientation, color, and shading. Relative positions (e.g., leftmost, center, top) are encoded explicitly, and the prompt requires exhaustive detail without summarization. Each image is converted to a data URI and dispatched with the prompt to a Gemini-2.5-based model, and the output is parsed into a structured bullet list via regex-based extraction. This structured description constitutes the perceptual anchor for the downstream agents, encoding both explicit and implicit spatial cues. The procedure retains subtle visual features, such as partial occlusions and minor shape variations, for subsequent relational and transformational reasoning.

Object Segmentation Agent. The object segmentation agent converts the verbose textual description into a discrete set of uniquely identified objects, disambiguating visually similar instances. Each object is assigned a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent applies a formatted prompt to the same Gemini-2.5 model to structure the description, and post-processing splits the output by object ID to enforce consistency. This representation reduces ambiguity in downstream attribute and relation extraction and enables precise mapping of complex scenes; by explicitly isolating objects, it also facilitates detection of overlaps, repetitions, and hierarchical groupings not evident in the raw description.

Attribute Extraction Agent. The attribute extraction agent maps each object to a structured JSON record encoding core properties: shape, size, color, shading, orientation, and positional descriptors. The prompt enforces strict JSON output, and responses are parsed via `json.loads()`, with fallback defaults for robustness against model inconsistencies. This encoding supports quantitative and qualitative object comparison for later reasoning stages. By preserving both absolute and relative measures, the agent maintains fidelity to the perceptual content while normalizing descriptors for computational consistency, and the JSON schema enables programmatic downstream operations such as symmetry computation and transformation detection.

Spatial Relation Agent. The spatial relation agent encodes object interactions as triples (ObjectA, Relation, ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments such as rows, grids, and repetition patterns. Consistent object IDs are enforced to preclude relational ambiguity, and parsing extracts output lines beginning with parentheses followed by automated consistency checks. This step transforms the static object description into a relational graph encoding spatial topology and hierarchy, enabling reasoning over both pairwise interactions and complex scene configurations, which is critical for symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. The grouping and symmetry agent abstracts higher-order patterns from the object set and spatial relations, identifying clusters, rows, grids, and symmetries (vertical, horizontal, rotational, and translational). Outputs are required in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to preserve downstream compatibility. The agent captures emergent scene regularities that are not evident at the object or pairwise-relation level, providing context for pattern recognition; by formalizing structural redundancies and symmetries, it supports both perceptual inference and anticipatory reasoning about dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the transformation tracking agent logs object-level changes, including translation, scaling, rotation, color/shading shift, and appearance or disappearance events. Structured logs are extracted line-by-line via regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding supports tracking of both static spatial configurations and dynamic evolution, enabling reasoning about motion, causality, and object interaction over time. The agent binds each transformation to a consistent object ID, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the representation standardization agent consolidates all outputs into a unified JSON record containing the keys `objects`, `relations`, `groups`, `symmetries`, and `transformations`. The prompt enforces consistent object IDs and structured formatting, and JSON parse failures revert to empty dictionaries to ensure pipeline robustness. By centralizing the perceptual, relational, and temporal information, this agent constructs a comprehensive scene representation that is both human-interpretable and machine-readable, serving as a unified foundation for downstream tasks including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation is passed to the reasoning agent, which answers multiple-choice or pattern-recognition questions. Its inputs comprise the objects, attributes, relations, groups, symmetries, and transformations, and it integrates geometric patterns, relational alignment, and dynamic transformations to generate or select the answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

The implementation uses Gemini-2.5-Flash at low temperature (0.1) to enforce approximately deterministic outputs and performs multiple refinement iterations. API calls are routed through OpenRouter, which supports both text and image inputs, and response parsing relies on regex and JSON-based extraction to produce structured outputs. All interactions are logged in JSON for reproducibility, and the final output is a single unified JSON record per image encoding the objects, relations, symmetries, and transformations. This multi-agent decomposition constrains each specialized model to a bounded sub-task, improving overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol is identical to the standard CoT and CoT-SR settings, using 5 class-balanced samples per subcategory (seed 42), for a total of 150 samples.

Table 25: Per-category accuracy of Gemini-2.5-Flash with and without the SPATIOXOLVER agents, under the MCQ and open-ended protocols, with the corresponding gains.

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents delineates both strengths and limitations of agentic reasoning on perceptual–geometric tasks. Orientation is the only category on which agents yield consistent and substantial gains, of +8.00% under MCQ and +36.00% under open-ended evaluation. This pattern indicates that agent-based multi-step reasoning is effective when the task decomposes into sequential alignment of directional cues, where structured deliberation offers a clear advantage over single-step outputs.

By contrast, most other categories exhibit negative or neutral gains, indicating that agentic reasoning does not universally improve accuracy. Depth & occlusion, for example, incurs a severe −24.00% decrement under open-ended evaluation, and spatial navigation decreases by −12.00%. These tasks require deeper perceptual grounding in three-dimensional space; absent such priors, the reasoning steps reinforce rather than correct misconceptions. Similarly, relative positioning stagnates under MCQ and declines by −14.00% under open-ended evaluation, indicating that agent loops do not add robustness when the underlying relational understanding is weak.

Some categories exhibit marginal improvements, such as size & scale (+4.00% under MCQ), but the gains are unstable and do not transfer to open-ended evaluation. Even 3D geometry does not benefit, with -4.00% and -12.00% decrements under MCQ and open-ended evaluation, respectively. These results establish that the benchmark is not solved by layering multi-step agentic reasoning atop existing models; it instead requires deeper integration of geospatial understanding and perceptual representations, which current architectures lack.

In aggregate, the mixed results indicate that agentic workflows alone are insufficient for advancing performance in this domain. Although orientation tasks benefit from structured reasoning, most categories expose the absence of internalized perceptual concepts, which causes errors to propagate through the multi-step pipeline and amplifies the decrements. Consequently, the benchmark functions as a strong stress test, indicating that future progress will require embedding genuine perceptual and geospatial priors rather than relying on agent-based reasoning alone.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section presents a focused qualitative analysis of the model failures observed on the benchmark. For each error class we provide a definition, a diagnostic root-cause analysis, the recurring patterns, and the associated verification protocols. The analysis is intended to be interpretable and actionable, so as to inform subsequent architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models exhibit strong coarse perception, reliably recognizing object categories, colors, and simple binary relations, yet they degrade systematically when multiple cues must be integrated. The failures are not random noise but partition into a small set of interpretable classes: spatial mislocalization, perspective and scale errors, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, indicating that the root causes are common inductive biases of current vision-language pipelines rather than isolated training artifacts. Notably, the same error frequently appears in both large-scale proprietary systems and small open-source baselines, indicating structural weaknesses shared across the model family. The following subsections characterize each error class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When a query requires distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. When asked to identify “the leftmost red cube among three stacked cubes,” for example, the model may return the middle cube, biased by lighting highlights or central positioning. This failure is most acute in cluttered environments such as kitchen counters or crowded table scenes, in which multiple overlapping objects share the same color and shape. In such cases the errors are not random: the model preferentially selects objects that are most salient by size, brightness, or centrality, even when these cues contradict the explicit spatial constraint in the query. This consistency indicates that the failures are systematic rather than incidental.

★ **Analysis.** The recurring errors indicate that the models lack a robust object-centric representation, i.e., a persistent index in the sense of cognitive science that is maintained across reasoning steps. Rather than maintaining a stable pointer to the correct candidate throughout the reasoning chain, the model drifts, reassigning attention to the most visually salient object at intermediate layers. This yields reasoning trajectories that begin at the leftmost cube but terminate at the brightest or largest cube. The failure to anchor the spatial referent consistently renders relative-comparison tasks, such as “the ball behind the second chair,” particularly fragile; in effect, the model operates as a salience detector rather than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature-pooling operations, whether attention- or convolution-based, mix contextual information across neighboring regions. Although this benefits global context, it erases the fine-grained, object-specific identity required to distinguish similar candidates: once pooled, the features of two adjacent red cubes are nearly indistinguishable in the latent space. Absent an explicit mechanism such as slot-based representations or pointer tokens, the

network collapses distinct entities into blended embeddings. This structural limitation precludes reliable referent tracking across reasoning steps and accounts for the systematic confusion of dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks requiring metric reasoning about relative size, fit, or scale are handled inconsistently. On simple comparisons such as “is a car larger than a bicycle,” models typically succeed because the relation aligns with a strong statistical prior. On near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” however, errors become frequent and unpredictable. A common failure is reliance on canonical object sizes: predicting, for example, that a door must be taller than a truck even in a rendering in which the truck is scaled down. A related failure occurs in atypical visualizations, such as toy miniatures or architectural models, in which familiar proportions are deliberately inverted; here the model adheres to its priors and disregards the geometric evidence in the image.

★ **Analysis.** This behavior indicates that models rely predominantly on the distributional statistics of object categories rather than performing geometric inference. In place of reasoning over projection geometry, relative bounding-box size, or depth cues, they substitute their prior over typical scales. This heuristic performs well in everyday contexts but fails catastrophically on cases outside the training distribution: when an unusually small airplane is rendered next to a person, the model still predicts the airplane to be larger. Such errors localize the absence of grounding in pixel-level metric cues and the lack of consistent use of perspective or vanishing-point geometry.

⊗ **Root cause.** Current architectures lack explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object-category labels and size rather than the underlying geometry. Moreover, the training objective optimizes aggregate accuracy rather than fine-grained metric consistency, so scale-sensitive errors are under-penalized. Absent inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors over canonical sizes instead of interpreting the perspective geometry present in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front of, behind, or partially occluding another remains a core weakness. Models frequently misreport the foreground object when two animals overlap or when semi-transparent barriers such as glass or fences are present; in an image where a cat sits partially behind a curtain, for example, the model may report the curtain as occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are especially error-prone and are frequently omitted entirely, yielding incorrect answers to queries such as “what object lies behind the fence.” These errors are systematic, particularly when the occlusion cues are subtle or the depth difference is small.

★ **Analysis.** These failures indicate that models treat depth and occlusion as weak secondary signals. Although the pixel data contain clear occlusion cues (e.g., edge boundaries and T-junctions), the intermediate features do not retain them at sufficient resolution: aggregation layers blur boundaries and down-weight thin structures, effectively removing the cues required to resolve ordering. This accounts for models succeeding on gross depth distinctions (foreground vs. background) while collapsing on fine-grained cases that require inferring partial occlusion. In practice they guess rather than infer, with a systematic bias toward treating the visually larger or brighter object as the foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and provides no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are readily lost when features are pooled over large receptive fields, and the training data under-represent subtle occlusion cases, particularly those involving transparency or fine boundaries. Consequently, the learned representations flatten layered scenes into 2D appearance maps, precluding robust depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (color, texture, material) with functional or semantic roles. A representative case is misjudging whether a crosswalk is safe: the

model may base its answer solely on the presence of painted stripes while ignoring the traffic-light state or approaching vehicles. Similarly, a surface with a metallic texture may be classified as “steel” even when it is painted plastic, and in tool-use contexts the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. Such confusions are frequent on tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors indicate that language priors strongly bias the predictions when the visual input is ambiguous. The decoder favors plausible narratives grounded in training-data co-occurrence, e.g., “crosswalks are safe when stripes are visible” or “shiny textures correspond to metal.” This heuristic yields semantically coherent answers even when the relevant visual evidence is not inspected, producing superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios; in effect, the model overfits to textual stereotypes rather than grounding in perceptual detail.

⊗ **Root cause.** The training objective, maximizing the likelihood of fluent and plausible text, does not enforce visual grounding. A semantically reasonable answer minimizes the loss even when the reasoning is incorrect, so the model is incentivized to substitute common-sense priors for perceptual inference. Absent auxiliary losses or verification steps that bind textual claims to image regions, models will continue to interchange perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** On explanatory or why-style questions, models frequently generate coherent but unsupported narratives. When asked “why is the object leaning,” for example, a model may answer “because the surface is slippery” even though slipperiness is not visually inferable, or explain a tilted shelf as “due to an earthquake,” a claim outside the image evidence. These errors reflect a tendency to fill evidential gaps with fluent but ungrounded rationales that misrepresent the actual evidence.

★ **Analysis.** The failure arises from a decoupling of text generation and visual grounding: the model produces high-quality, convincing language, but no mechanism binds the rationale to the actual visual features. Confidence miscalibration compounds the problem, as the model frequently assigns high certainty to fabricated explanations, creating a false impression of reliability. Consequently, open-ended outputs can appear authoritative while being factually baseless, which is particularly consequential in safety-critical domains where a plausible but false rationale could misdirect decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence rather than factual grounding. The decoder is optimized to produce the most probable text continuation given the prompt, with no constraint linking the explanation to specific image evidence. Absent an explicit verification layer or grounding requirement, the model defaults to linguistic plausibility over perceptual truth, and open-ended rationalization will continue to prioritize surface coherence over evidential accuracy until additional architectural or training constraints are imposed.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

A cross-model comparison of failures reveals several consistent patterns that characterize the underlying error mechanisms. First, inter-model disagreements are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relations, whereas others default to linguistic priors or common-sense assumptions from training text. This yields predictable divergences, e.g., two models returning different answers to which object is “in front” of a shared scene. Second, the error rate increases sharply when multiple cues must be integrated simultaneously; tasks requiring joint reasoning over occlusion, relative scale, and perspective are particularly difficult, and determining whether a larger object partially occluding a smaller one is in front or behind consistently triggers mispredictions. Third, calibration is degraded: high-confidence predictions frequently correspond to incorrect answers, particularly in open-ended explanations and complex spatial queries. These patterns indicate that the errors arise from how models balance perceptual grounding against linguistic plausibility, and that different architectures exhibit distinct tendencies, some over-relying on visual geometry and others overfitting to language correlations, so the failures reflect deeper trade-offs in reasoning strategy rather than isolated mistakes. This implies that improving performance requires addressing both sources of bias jointly rather than treating the errors as independent.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can reduce failures without a complete architectural redesign. First, geometry-aware supervision can be introduced via auxiliary tasks such as depth prediction, occlusion ordering, or overlap estimation, which enforces spatial constraints and improves cross-prediction consistency. Second, multi-scale feature retention, via edge-aware pooling or attention, preserves information from thin or partially occluded structures that would otherwise be lost under standard downsampling. Third, coordinate anchoring can resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Fourth, counterfactual augmentation, in which distractor objects are systematically swapped or perturbed, forces the model to rely on the true referent cues rather than heuristics. Fifth, lightweight verification modules can be appended to the existing outputs to cross-check claims against visual or geometric constraints and filter implausible predictions. Each strategy is incrementally deployable, providing a concrete path to reducing systematic failures while leaving the core model largely unchanged.

Beyond these architectural and training augmentations, supervised fine-tuning (SFT) provides a practical route to more reliable spatial reasoning. By curating datasets that explicitly instantiate failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference-resolution traps, SFT directly exposes the model to the edge cases on which it otherwise defaults to heuristics. Instruction-style fine-tuning further promotes step-by-step reasoning, requiring the model to verbalize intermediate spatial relations rather than emit a direct guess, which strengthens grounding and improves interpretability by making errors traceable to specific reasoning steps. SFT can additionally be paired with counterfactual or adversarial augmentation, so that the model learns invariance to distractors and context shifts rather than memorizing. Critically, these interventions require no wholesale architectural change: they reshape the inductive biases of existing models through supervision, so that SFT serves as a lightweight complement to geometry-aware and verification-based strategies.

In this work we additionally performed SFT on 40% of the training data and observed clear gains, most pronounced under the multiple-choice protocol. This indicates that targeted supervision helps the model internalize spatial cues and resolve distractor confusions when the answer options are well-defined. Improvements on open-ended responses were more limited, however: the model continues to produce fluent but weakly grounded explanations, indicating that SFT alone does not resolve rationalization absent verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentation, and lightweight verifiers to achieve balanced progress across the structured and free-form protocols.

Our analysis establishes that failures are not arbitrary but structured and interpretable, arising from recurring patterns rather than random noise. This indicates that VLM errors follow systematic trajectories tied to the inductive biases of the architecture and training regime. Geometric reasoning, for example, collapses when occlusion cues are weak, whereupon language priors dominate and produce plausible but visually ungrounded answers; likewise, scale inference drifts when canonical object-size priors dominate the pixel-level metric cues, yielding consistent misjudgment in near-threshold scenarios. These findings indicate that the gap is one of representation rather than capacity: current models lack explicit mechanisms to bind visual features to grounded geometric relations.

Adopting the proposed verification protocols and targeted mitigation strategies addresses these weaknesses directly. Geometry-aware supervision and counterfactual augmentation help the model resist shortcuts, and multi-scale feature retention ensures that thin structures and partial occlusions are not discarded. SFT, when carefully applied, further strengthens structured reasoning by aligning predictions with the task constraints, but it must be combined with grounding objectives to avoid overfitting to discrete-choice formats. Verification modules provide a lightweight mechanism for enforcing consistency between claimed relations and image evidence, counteracting the tendency to rationalize without verification. Collectively, these steps constitute a practical path toward systems that reason more like humans: robust to clutter, interpretable in their failures, and able to explain not only what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position":
 "Leftmost" }},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position":
 "Center" }}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting

3. Example Output:

```
{  
  "objects": [...],  
  "relations": [...],  
  "groups": [...],  
  "symmetries": [...],  
  "transformations": [...]}  
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark items for all 30 subcategories of SPATIALAB, illustrating the range of spatial reasoning challenges covered. These items span the visual phenomena of 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation, and together they concretely specify the reasoning that a model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- Gemini-2.5-Flash: The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- InternVL3.5-72B: The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- Kimi-VL-A3B-Thinking-2506: Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

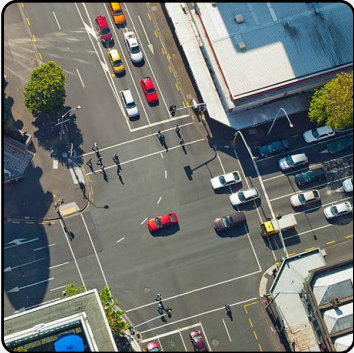
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

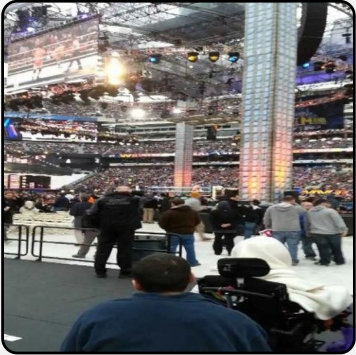
Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini:** 3
- Gemini-2.5-Flash:** 3
- InternVL3.5-72B:** 4
- Kimi-VL-A3B-Thinking-2506:** 4

Open-ended Answers:

- GPT-5-mini:** If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash:** The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B:** If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506:** The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

To establish that the benchmark results are not artifacts of random initialization or dataset noise, we quantify both model robustness across runs and dataset stability across repeated trials. Such analyses are essential for small- to medium-scale benchmarks, in which minor fluctuations can yield misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items (questions) in the benchmark and $R = 3$ the number of independent runs of a given model. Each run uses a distinct random seed while holding all other hyperparameters fixed.

K.1 MODEL ROBUSTNESS

To quantify the robustness of the model evaluation, we conduct the statistical studies described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify the agreement between runs, we compute the intra-class correlation coefficient (Shrout & Fleiss, 1979) under a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ denotes the between-item variance and σ_{within}^2 denotes the residual variance across runs. ICC values above 0.75 denote good reliability, and values above 0.9 denote excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We further analyze the stability of the dataset itself, namely whether item difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further quantify **robustness**, we perform a resampling analysis. From the $C = 30$ subcategories, we draw random subsets of size $S \in 20, 25$ per subcategory from each of the $R = 3$ runs, for both models and both evaluation protocols (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, we apply the Wilcoxon signed-rank test to assess median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ denote the accuracy difference for run r , excluding ties ($d_r \neq 0$). The test statistic is

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we fail to reject H_0 , which confirms that the accuracies from the 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty is defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the inter-run correlation should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) indicates stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To quantify overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 denotes the variance of run r and s_T^2 denotes the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ denote excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We conduct the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby covering both the large- and small-scale regimes and the proprietary and open-source paradigms. All tests are conducted under both the multiple-choice (MCQ) and open-ended protocols, with each response treated as binary (correct or incorrect). For the statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for wilcoxon test; `pingouin`³ (Vallat, 2018) for intraclass correlation and cronbach alpha; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We quantified the stability of model accuracy across $R = 3$ independent runs. The across-run standard deviation of accuracy (σ_A) is negligible ($< 0.6\%$), indicating that the reported scores are precise point estimates that are unaffected by random-seed noise.

Table 26: Model accuracy and across-run variance over 3 independent runs.

Model	Format	Mean Accuracy ($\bar{A}$)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and protocols indicate that the accuracy estimates are stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability indicates that the observed accuracy differences reflect underlying model capability rather than noise from random initialization or decoding. Although the open-ended protocol exhibits marginally higher variance than MCQ, the absolute magnitude remains negligible, confirming that free-form generation is also robustly assessed under the current protocol. These results indicate that the dataset size and scoring methodology suffice to yield convergent accuracy estimates without extensive repeated runs; consequently, single-run evaluation is reliable for large-scale comparison, and multi-run analysis primarily serves to validate robustness. In aggregate, this strengthens the empirical credibility of the reported results and supports confident cross-model and cross-protocol interpretation.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-class correlation (ICC) reliability scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores establish near-perfect reliability across independent runs, independent of model size or response format. This indicates that the item-level difficulty ordering is preserved and that accuracy fluctuations do not alter the relative rankings among items. Such strong agreement indicates that the scoring pipeline is internally consistent and insensitive to stochastic variation in the model outputs, and it rules out random guessing as a dominant factor even under the

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

open-ended protocol, where variability is typically higher. From an evaluation standpoint, these results indicate that the benchmark provides a dependable signal of model capability, so that longitudinal comparisons and fine-grained ablations can be conducted with high confidence in measurement stability.

In aggregate, the combined evidence from the multi-run variance analysis and the ICC reliability establishes that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) are stable across independent runs, confirming that the results are not driven by random seeding or decoding noise. This robustness ensures that the observed performance gaps across models and protocols reflect genuine capability differences, so that the evaluation framework provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, the resampling procedure is repeated over **1,000 independent trials**, each drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that the conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling study (1,000 trials): Wilcoxon signed-rank test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide strong empirical evidence that the accuracy estimates have converged at relatively small subcategory sizes. The consistently high mean p -values and the absence of statistically significant differences across all 1,000 trials indicate that the accuracy estimates are insensitive to moderate reductions in sample size. This stability holds uniformly across models and protocols, indicating that neither model scale nor output structure introduces additional sampling uncertainty. These findings indicate that the variance in accuracy is dominated by model behavior rather than by data sparsity at the subcategory level. Consequently, the current subcategory size (≈ 50) is statistically redundant, providing a substantial margin beyond the minimum required for reliable estimation. This supports the validity of the benchmark design and indicates that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We computed the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s alpha to quantify internal consistency.

Table 29: Item-level consistency metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The very high mean pairwise correlations (> 0.94) establish strong item-level stability, indicating that the relative difficulty of individual questions is preserved across independent runs. This consistency indicates that the model responses are systematically structured rather than subject to random variation, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) further confirm excellent internal coherence, with items contributing reliably to the aggregate accuracy measure. The slight reductions observed under the open-ended protocol are expected, given its greater expressive flexibility, yet remain within the range of excellent reliability. In aggregate, these results establish that the benchmark functions as a unified and internally consistent measurement instrument, so that

the observed performance differences can be confidently attributed to true capability variation rather than item-level instability.

K.3.3 OVERALL DISCUSSION

Our expanded statistical testing establishes that the evaluation and dataset are robust: the **ICC and Cronbach's α scores consistently exceed 0.98**, and the **across-run standard deviation of accuracy is negligible**. Furthermore, the **resampling study** confirms that smaller subsets produce accuracy estimates statistically indistinguishable from the full set, establishing that the subcategory sizes (~ 50) are sufficient for robust conclusions. These results establish that the fine-grained analysis reported in the paper reflects genuine differences in model capability rather than statistical noise.

Our expanded statistical testing establishes that the evaluation protocol and dataset are highly robust, with **ICC and Cronbach's α scores consistently exceeding 0.98**, indicating near-perfect reliability and internal consistency. The **across-run standard deviation of accuracy remains negligible** ($< 0.6\%$), confirming that the reported model performance is stable and not influenced by stochastic effects such as random seeding or decoding variability. In addition, the **resampling study** demonstrates that substantially smaller subsets ($N = 20$ and $N = 25$) yield accuracy estimates statistically indistinguishable from the full set, establishing that the chosen subcategory size (~ 25) exceeds the convergence threshold by a wide margin. Collectively, these results establish that both the absolute scores and the relative rankings are stable across runs and sampling regimes. This statistical robustness ensures that fine-grained comparisons across models, protocols, and subcategories are meaningful, so that the granular analyses reported in the paper reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation noise.